

Comparative genomics using reference genomes

Introduction to
Biodiversity Genomics
Mendoza, Argentina
2025



Comparative Genomics

NHGRI FACT SHEETS
genome.gov

Researchers choose the appropriate time-scale of evolutionary conservation for the question being addressed.

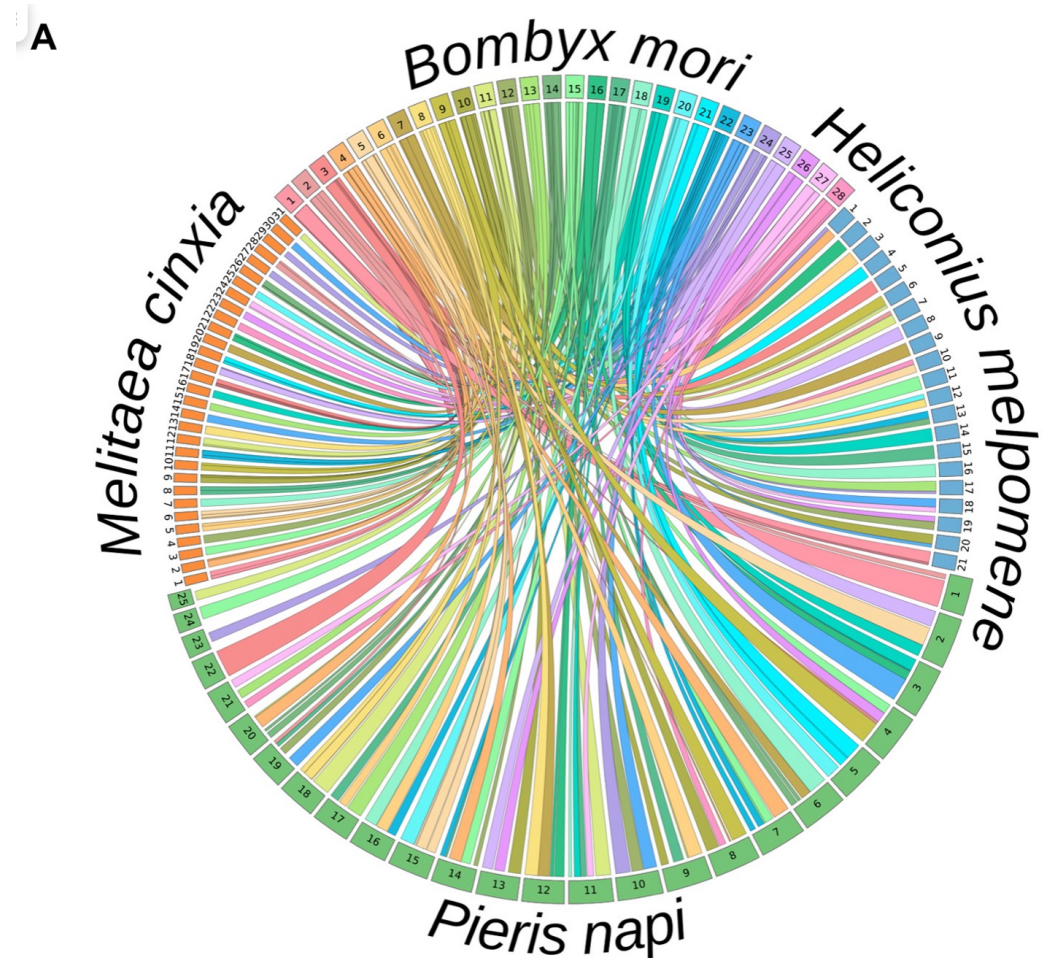
Common features of different organisms such as humans and fish are often encoded within the DNA evolutionarily conserved between them.

Looking at **closely related species** such as humans and chimpanzees shows which genomic elements are unique to each.

Genetic differences **within one species** such as our own can reveal variants with a role in disease.

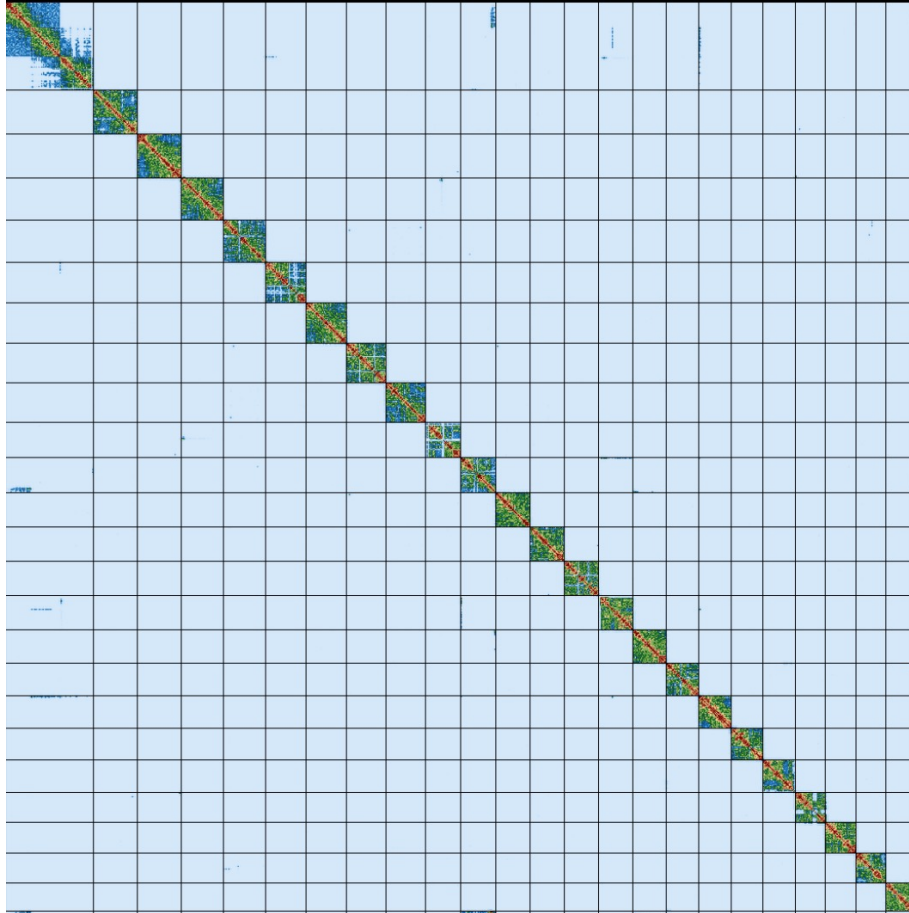
Overview of comparative genomics

- Signs of selection dN/dS
- Genome structure - size
- Genome features
 - Gene and regulatory dynamics
 - Gene family dynamics
 - Conserved non-coding elements
 - TEs and repeats
 - CG-content, codon usage and tRNA dynamics
- Heterozygosity
- Genome structure - karyotype



Genome structure

- size



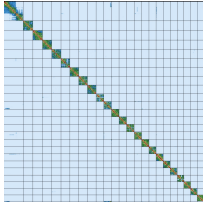
Melinaea menophilus zaneka



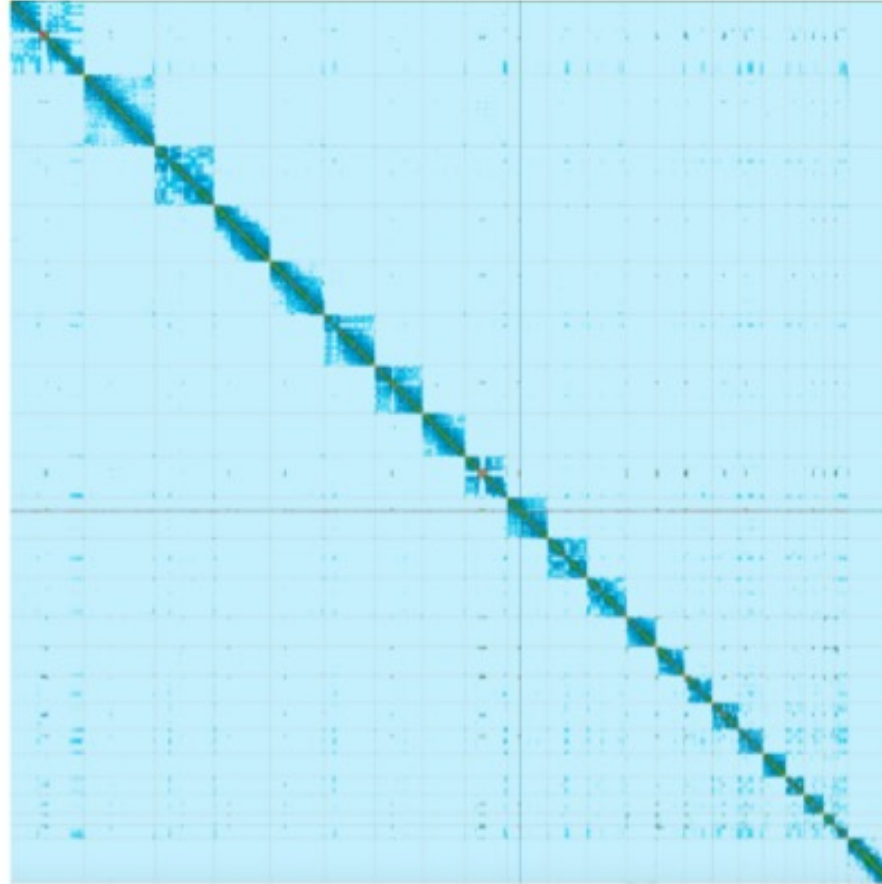
679 Mb

Genome structure

- size



600Mb



Homo sapiens, 3.1 Gb



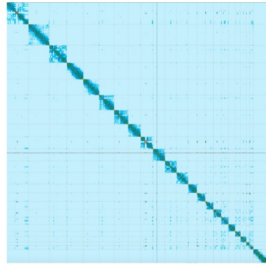
Donaldson collection/getty images

Genome structure

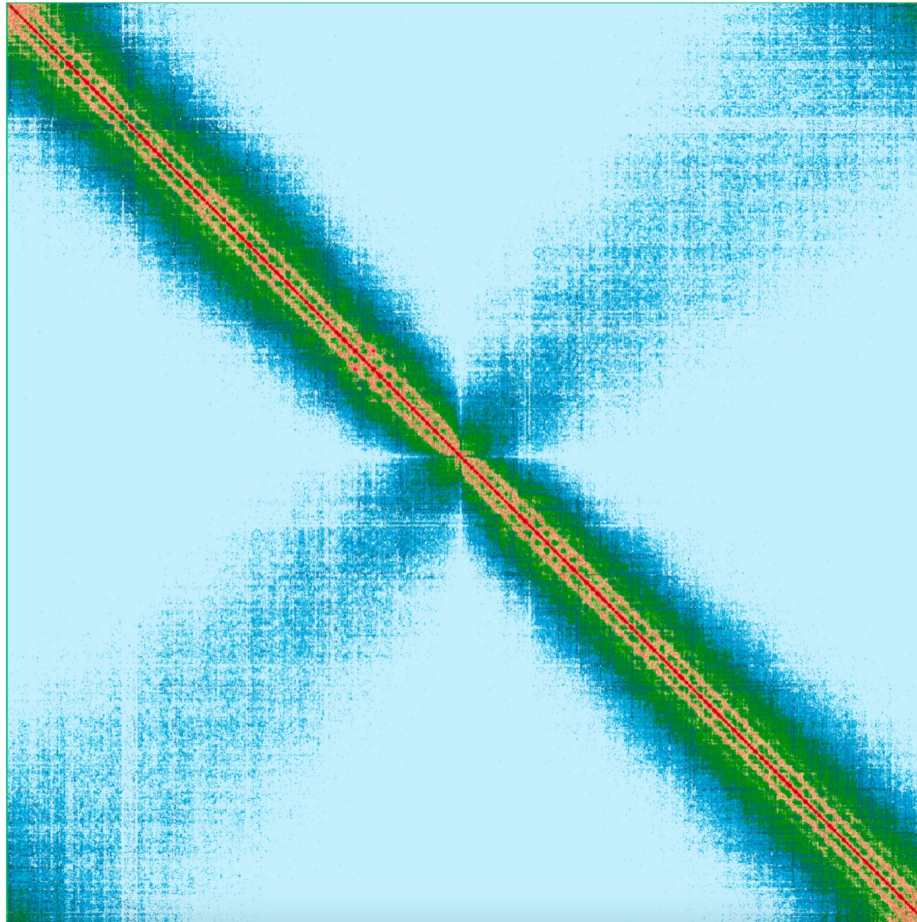
- size



600Mb



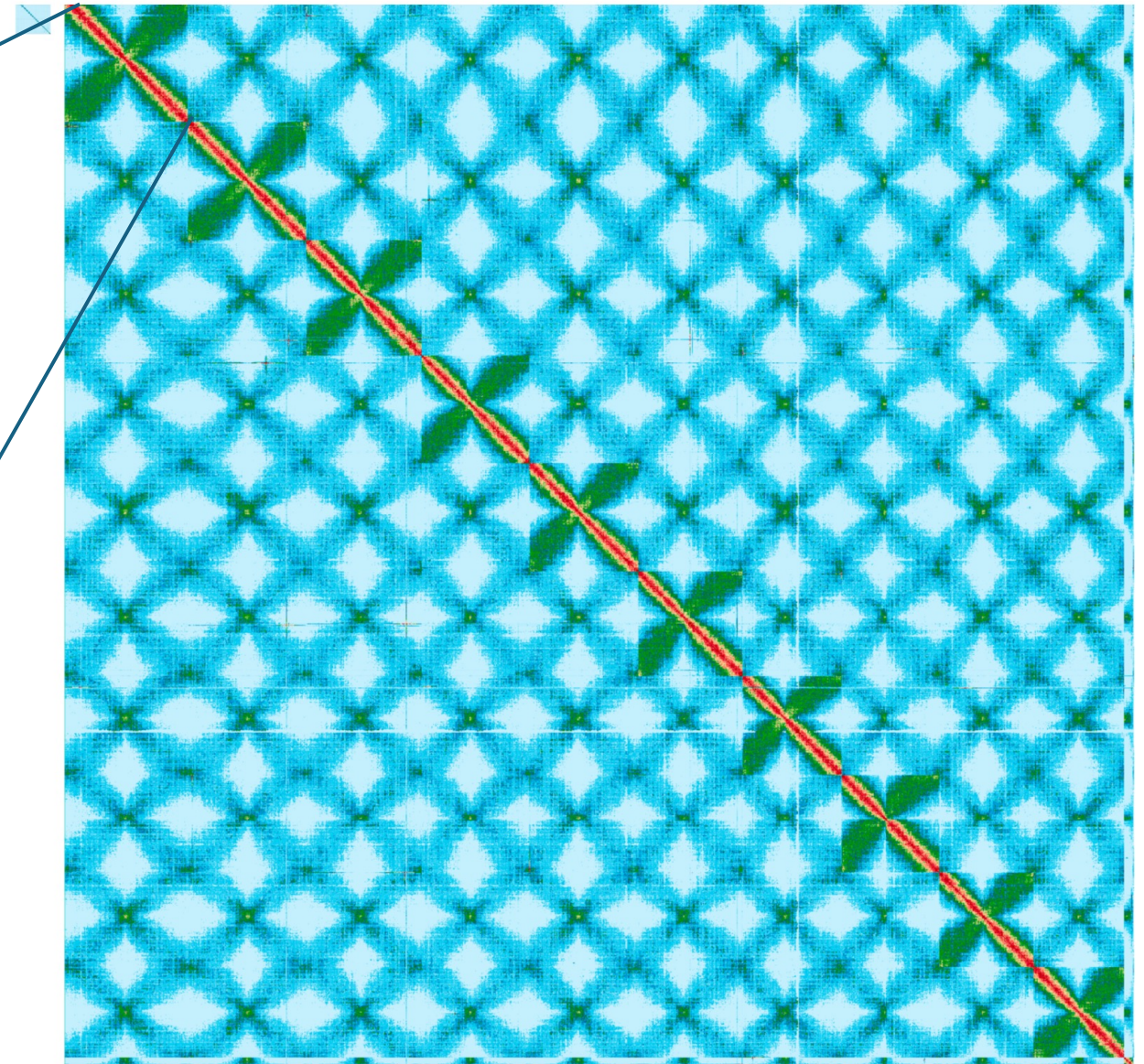
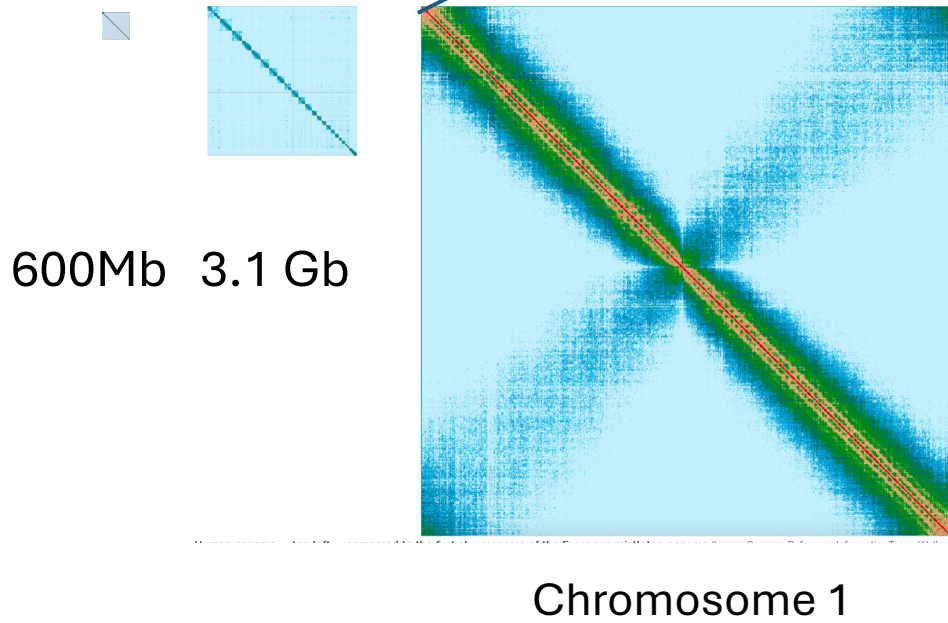
3.1 Gb



Mistletoe
Chr 1 ~11Gb

Genome structure - size

Viscum album 94.3 Gb



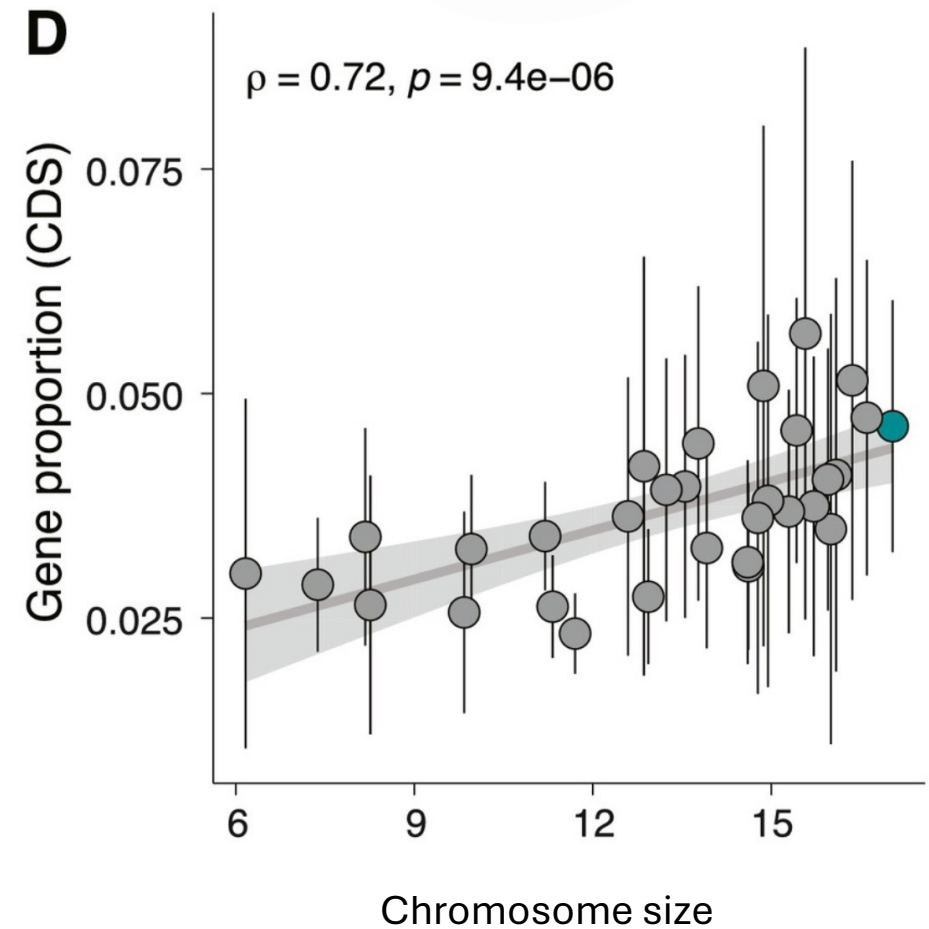
Human genome — again, top left — compared to the entire European mistletoe genome (Image: Genome Reference Informatics Team, Wellcome Sanger Institute)

Genomic features

- Gene family expansion and contraction
- Regulatory element evolution
- Transposable element dynamics
- Genome composition evolution

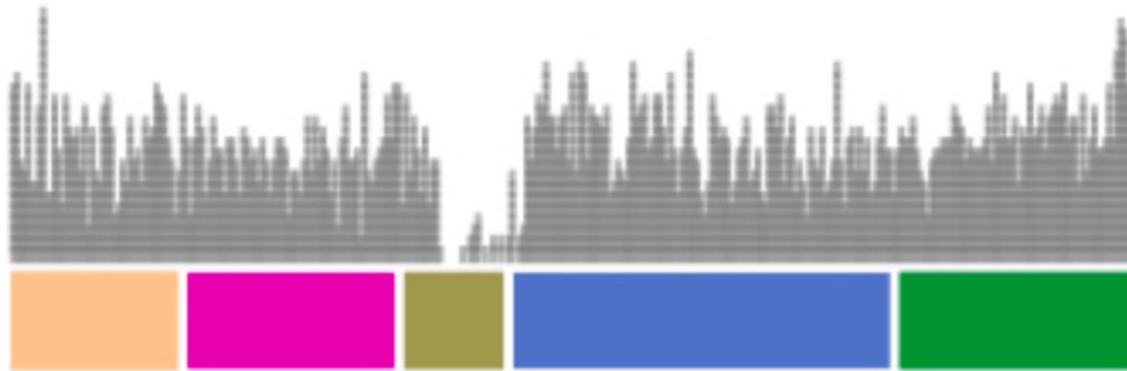
Gene content

- Protein coding genes and non-coding RNA within the genome
- Content vary
 - across species
 - among individuals
 - along genome



Gene content across the genome

Gene density



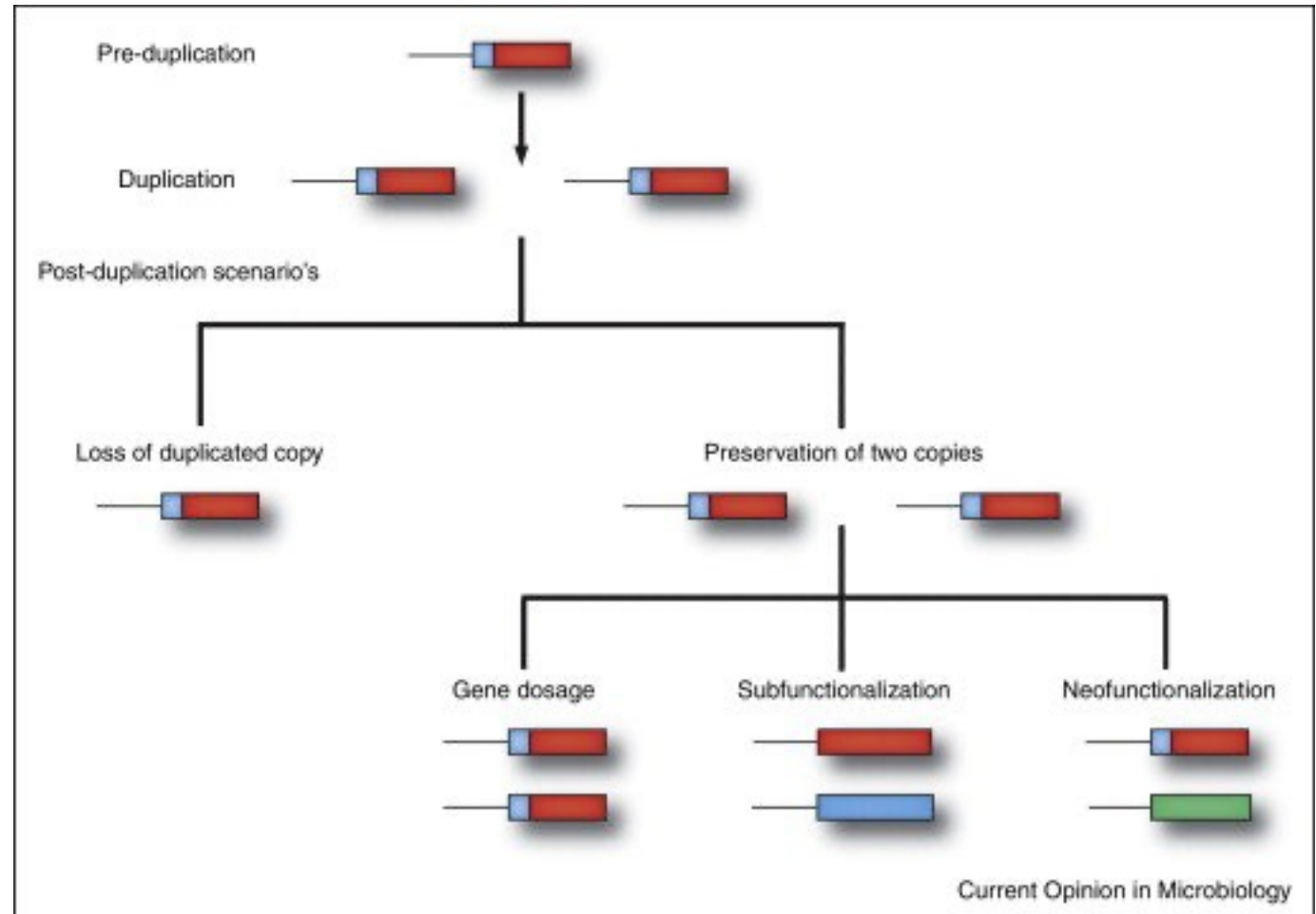
Chr 8



Arachis duranensis
Ren et al 2018

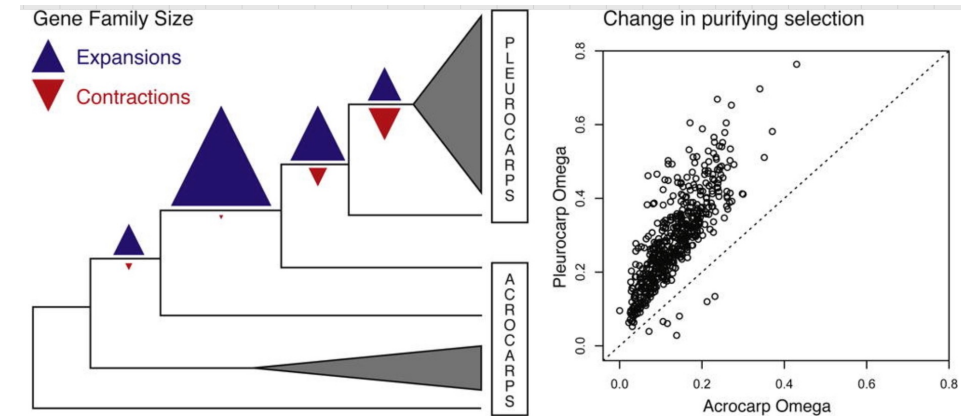
Gene duplications or deletions

- Gene gain or loss leads to structural variation between individuals
- Novel gene copies – adaptive potential



Gene family expansion or contraction

- Cluster genes in gene families by sequence homology
 - OrthoFinder
- Infer phylogenetically informed rates of gain and loss
 - Birth-death model for evolutionary inferences
 - Café, Badirate



Hypnum imponens (photo: B. Shaw)

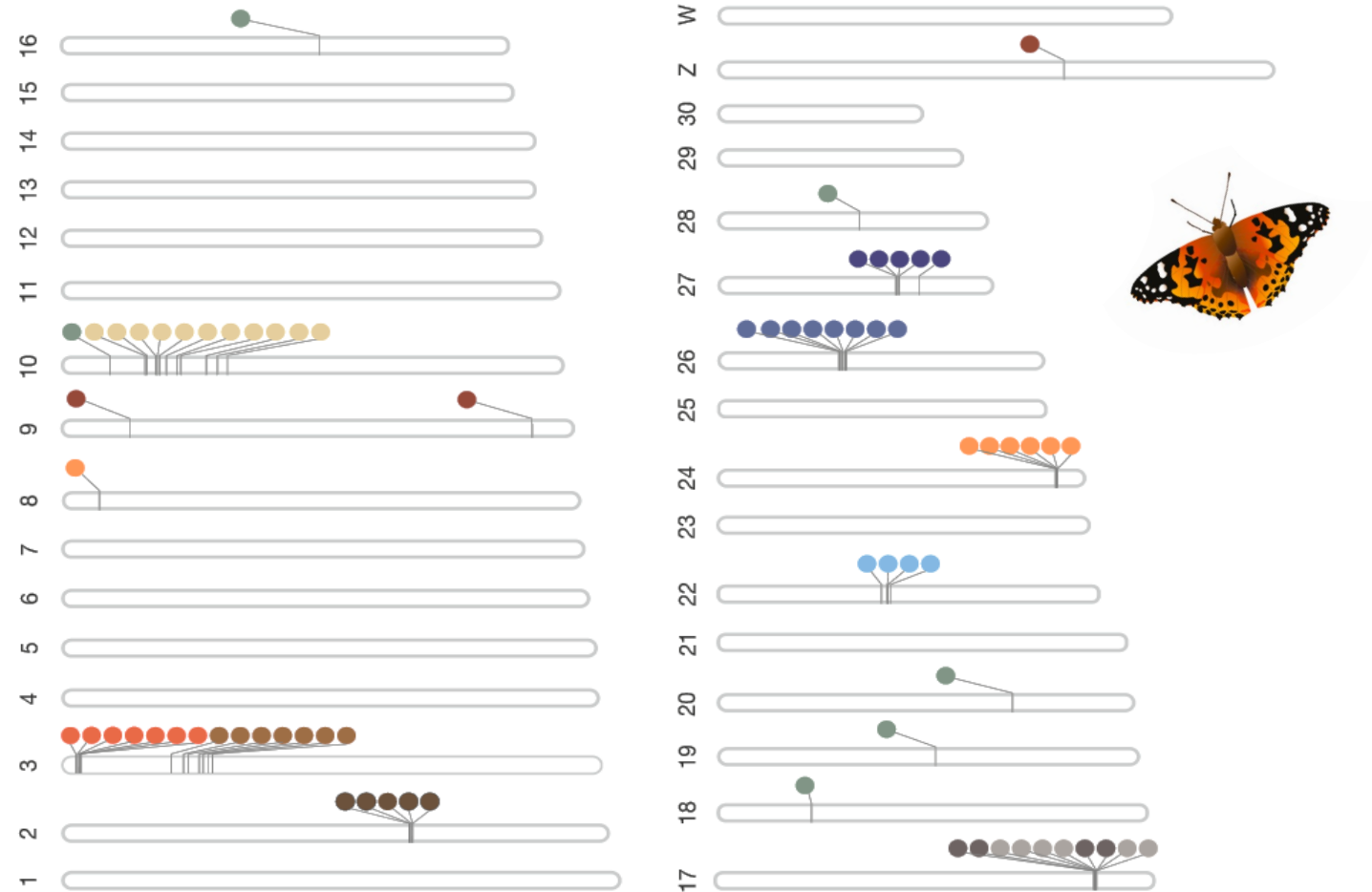
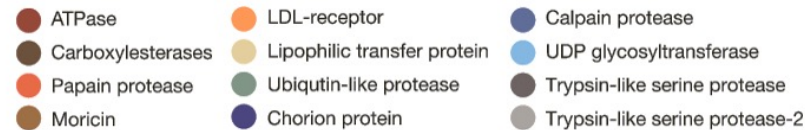


Hylocomium splendens (photo: B. Shaw)

Gene family expansion or contraction

- Gene ontology analysis, approximation of gene function
 - Panther, TopGO
 - Domain analysis
- Functional validation
 - Gene editing/knockout

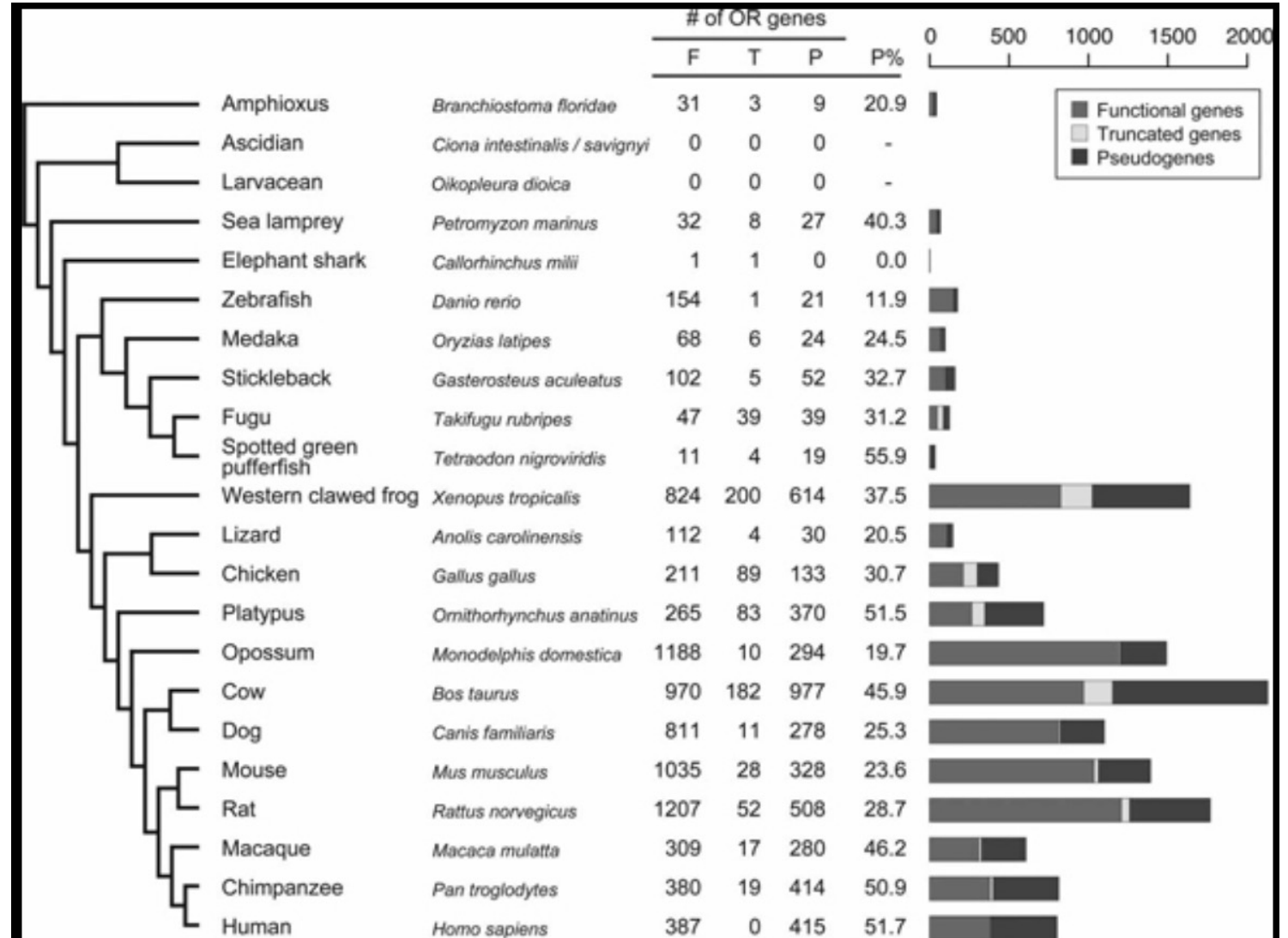
Candidate gene families



Gene family expansion or contraction

- Expansion of the olfactory receptor (OR) gene family
- Large variation
 - 1,200 functional copies in rats
 - 400 in humans
 - 15 in pufferfish

(Nimura, Hum Gen, 2009)



Regulatory elements

- Promoters binding site for polymerase
- Proximal and distal enhancers, silencers, insulators

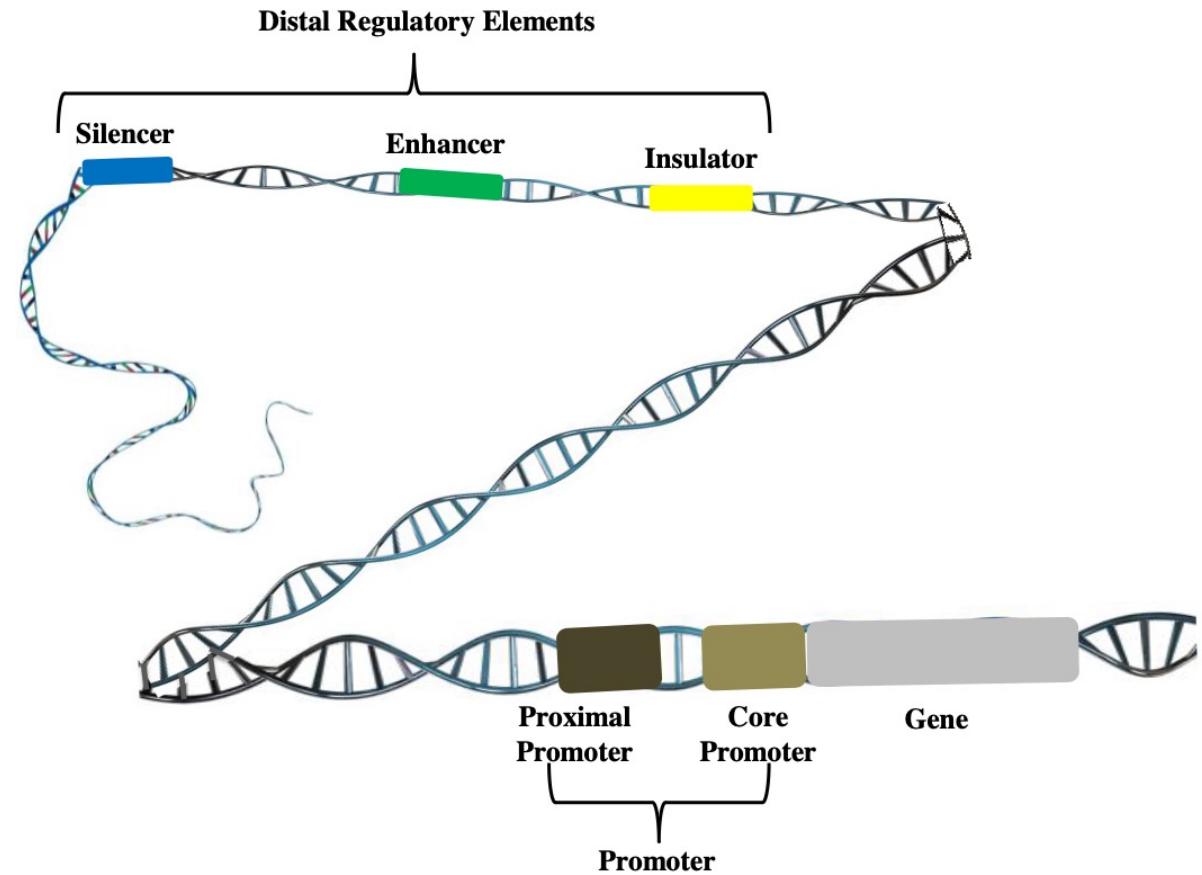
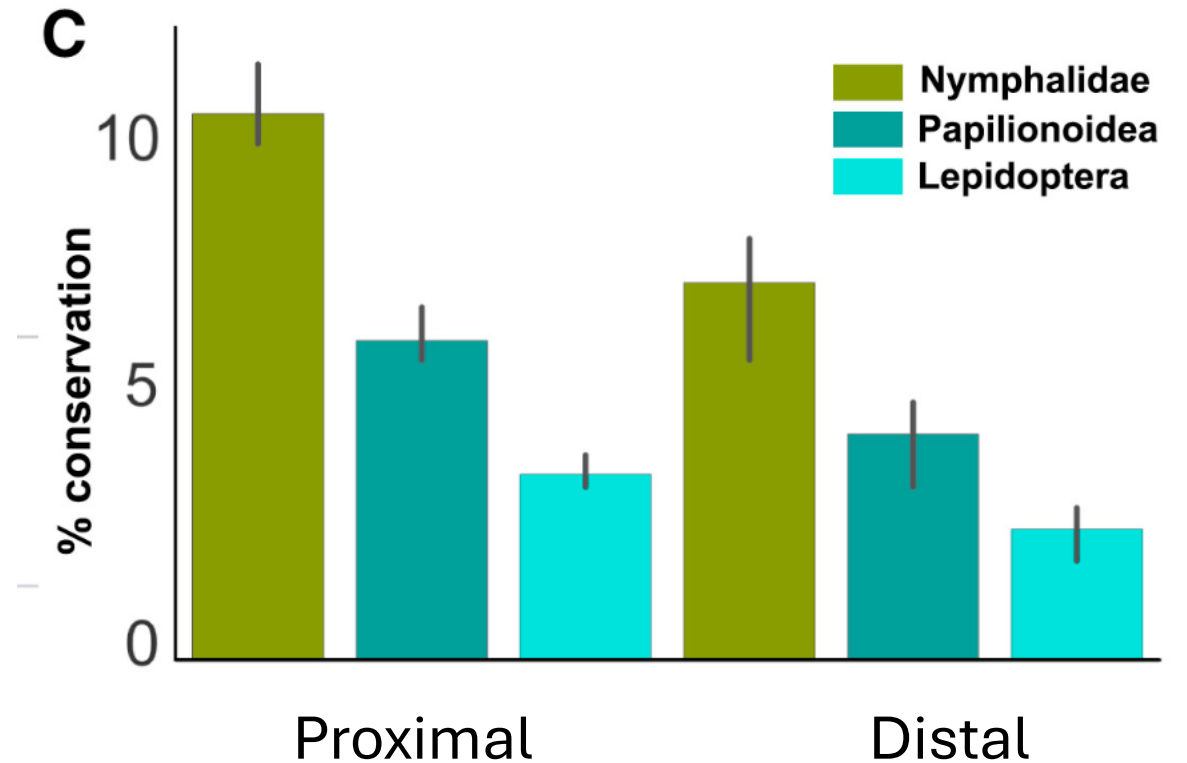


Fig. 1: Schematic representation of typical gene regulatory region.

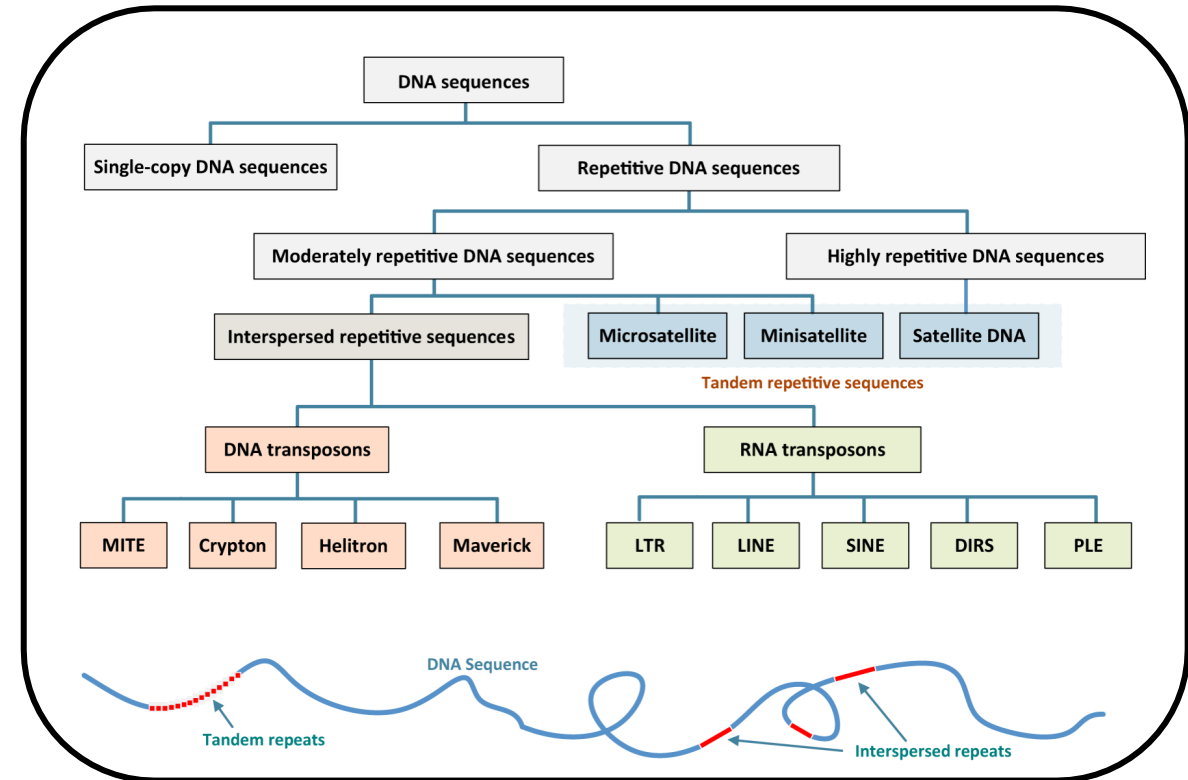
Regulatory elements

- Single nucleotide mutations, copy number variation (CNV) or TE insertions can change the regulatory function
- Infer by sequence homology
 - Whole genome alignment – conserved non-coding elements
 - Database search, known elements
 - Prediction tools, specific motif
- Predict promoter - enhancer interactions in HiC-data
 - PSYCHIC (Ron et al 2017)
- (Experimental approaches
 - ChIP-seq
 - ATAC-seq)

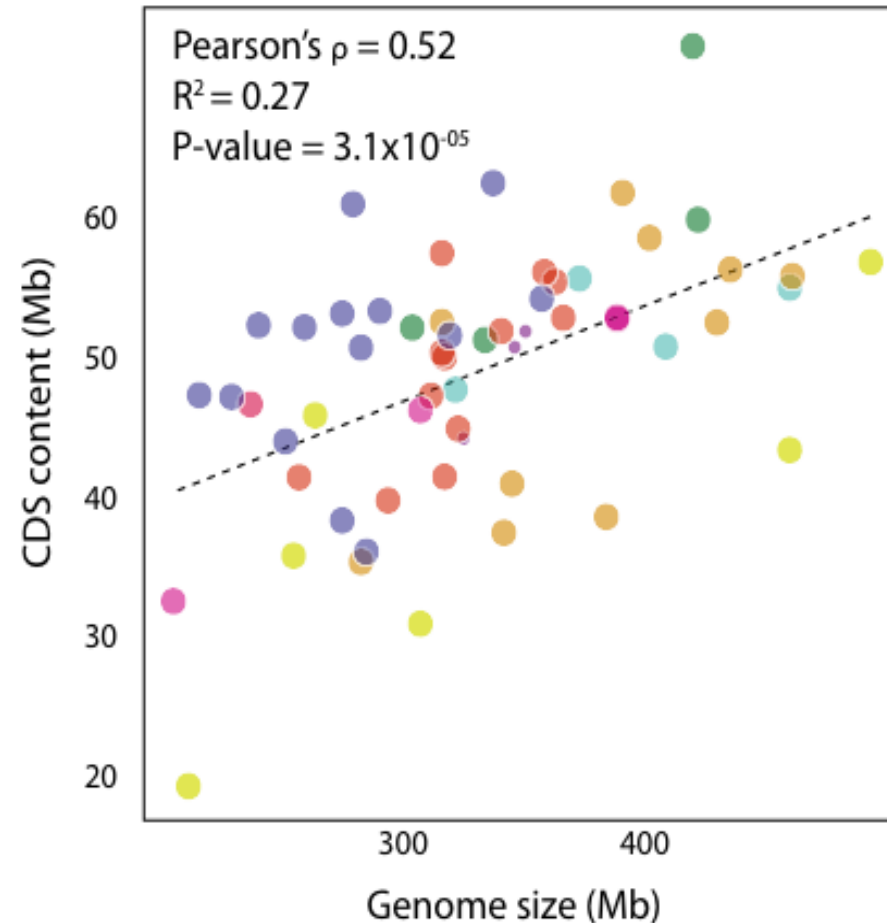
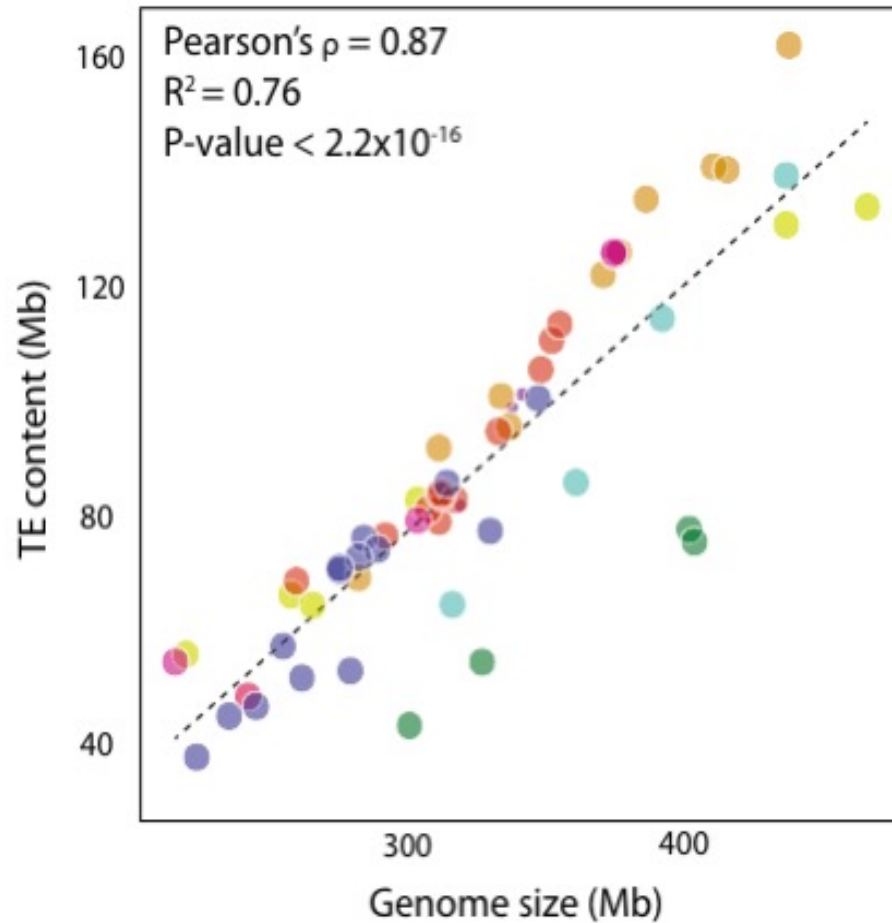


Repetitive sequences

- Transposable elements (mobile elements)
 - Move (or duplicate) from one location in the genome to another
 - Initiates their own transcription or retrotranscription and insertion using a wide range of endonucleases
- Simple repeats
 - satellite repeats
 - microsatellites
- Multi-copy RNA genes
 - rRNA, tRNA, snRNA

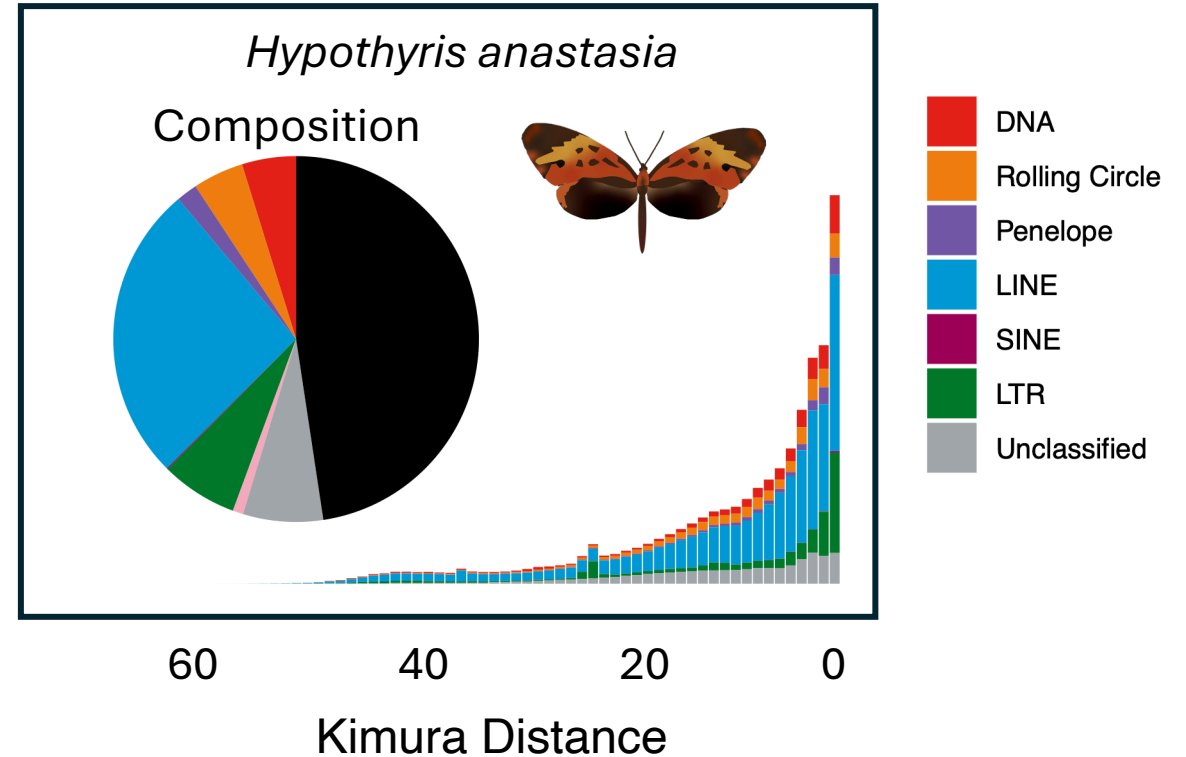
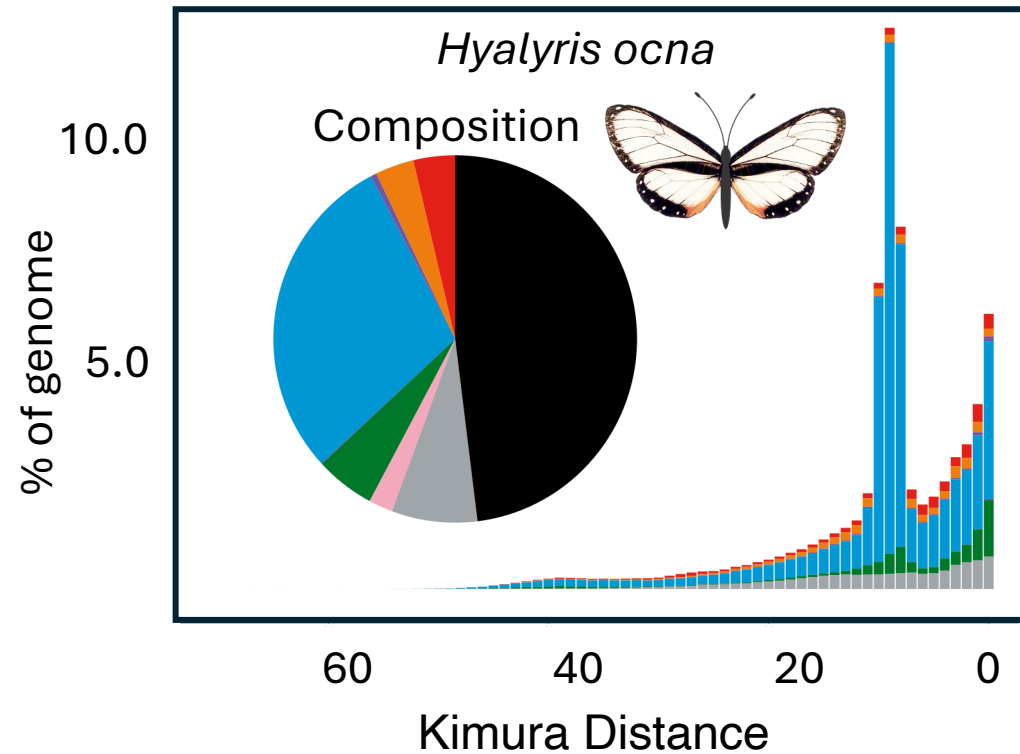
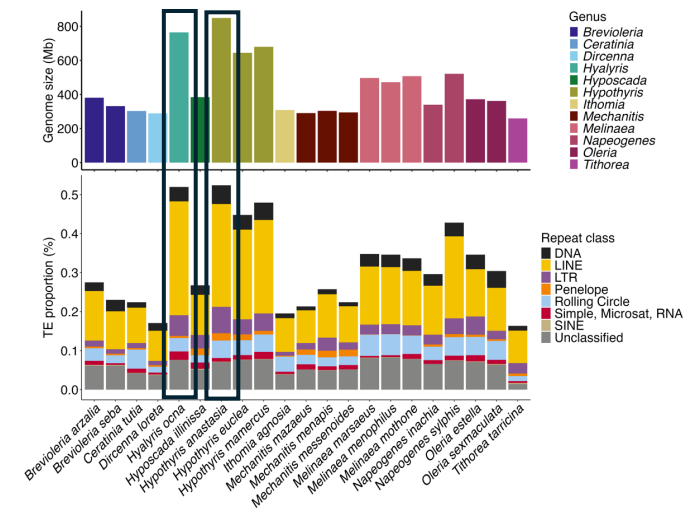


Transposable elements – genome size



Dynamics through time

- Infer TE divergence time



Transposable elements - annotation

- TE prediction algorithms using structure and sequence homology
- Consensus building and classification steps
 - RepeatModeler2 (includes the predictors above)
- Manual curation
- Annotation tools: EDTA, EarlGrey, Pantera

Satellite repeats

- Centromere-associated repeats
 - Hierarchical tandem repeat arrays
 - Annotate location of centromere
- Telomeric repeats
 - TeloBase
<https://doi.org/10.1093/nar/gkad672>
 - $(T_xA_yG_z)_n$
 - Ex, many insects $(TTAGG)_n$ $(TTCGGG)_n$
 - Detect differences in length or sequence shift

- Microsatellites

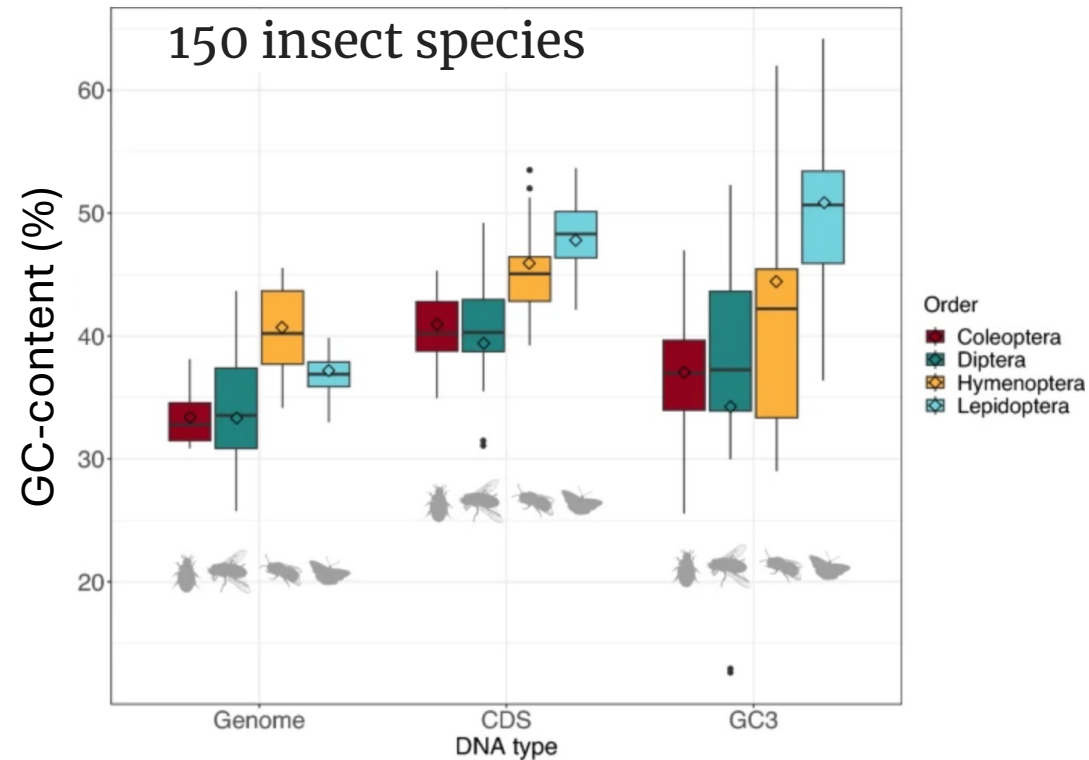
Ind1 ATATATATATAT

Ind1 ATATATATAT—

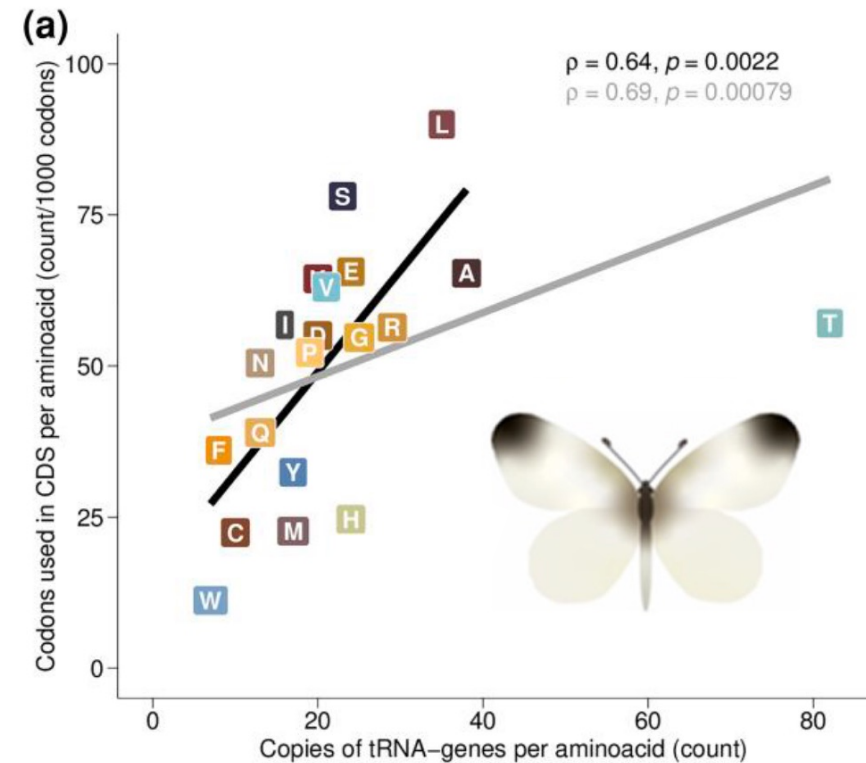
Ind2 ATATATATAT--

Ind2 --ATATATAT--

GC-content, codon usage and tRNA dynamics



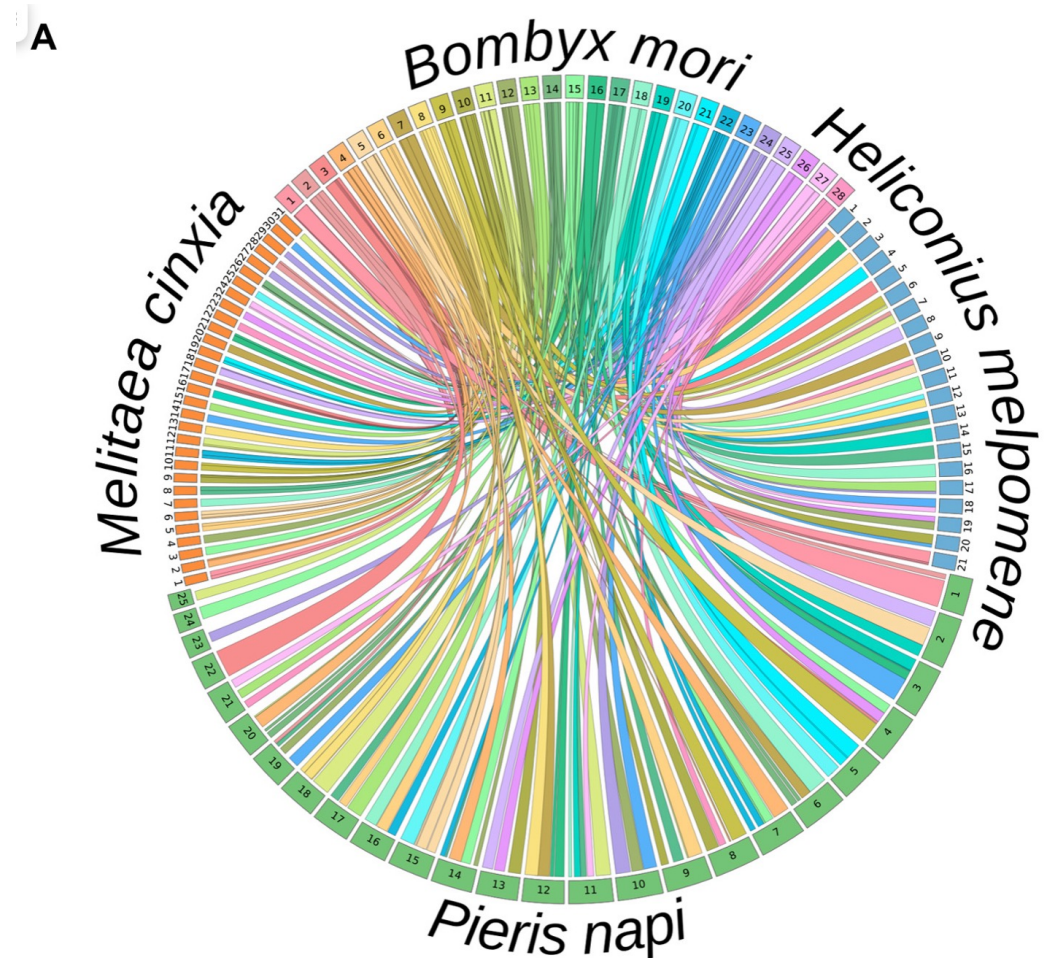
Kyriacou et al. 2024



Näsvall et al. 2023

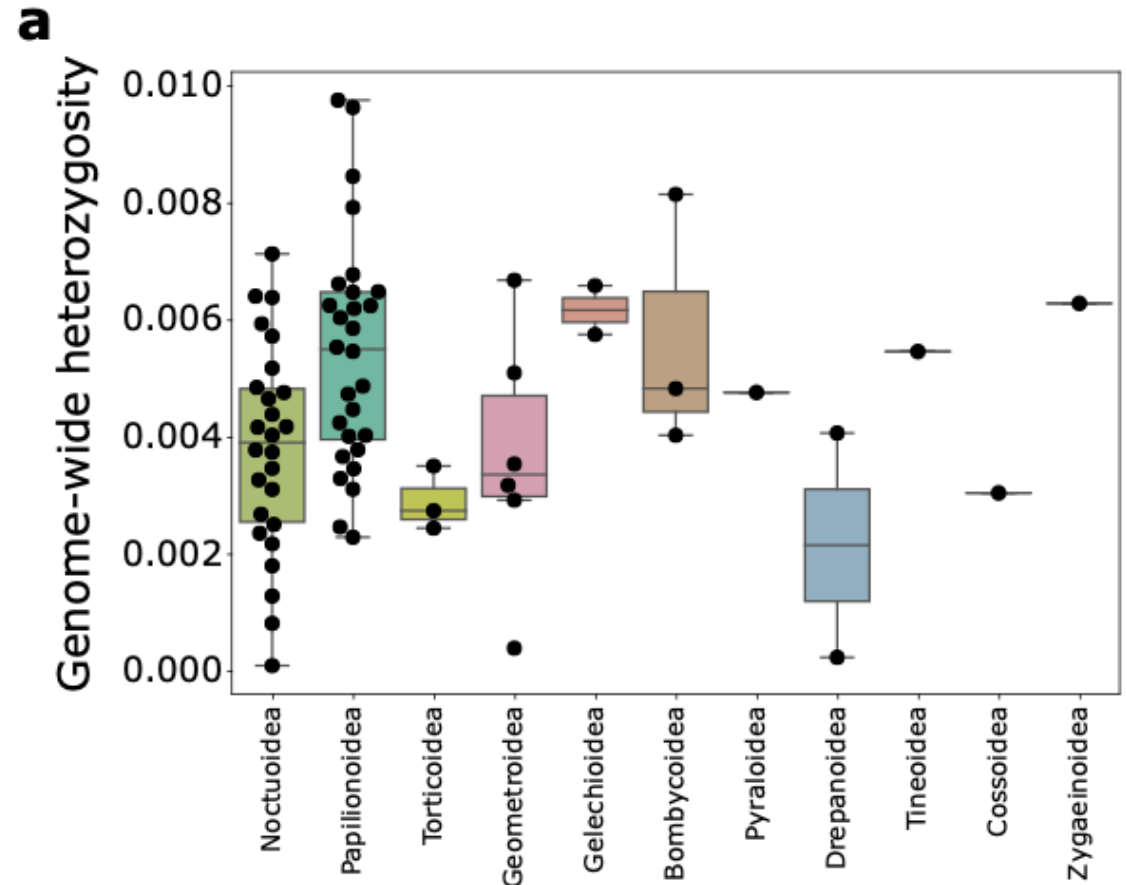
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- **Heterozygosity**
- Genome structure - karyotype



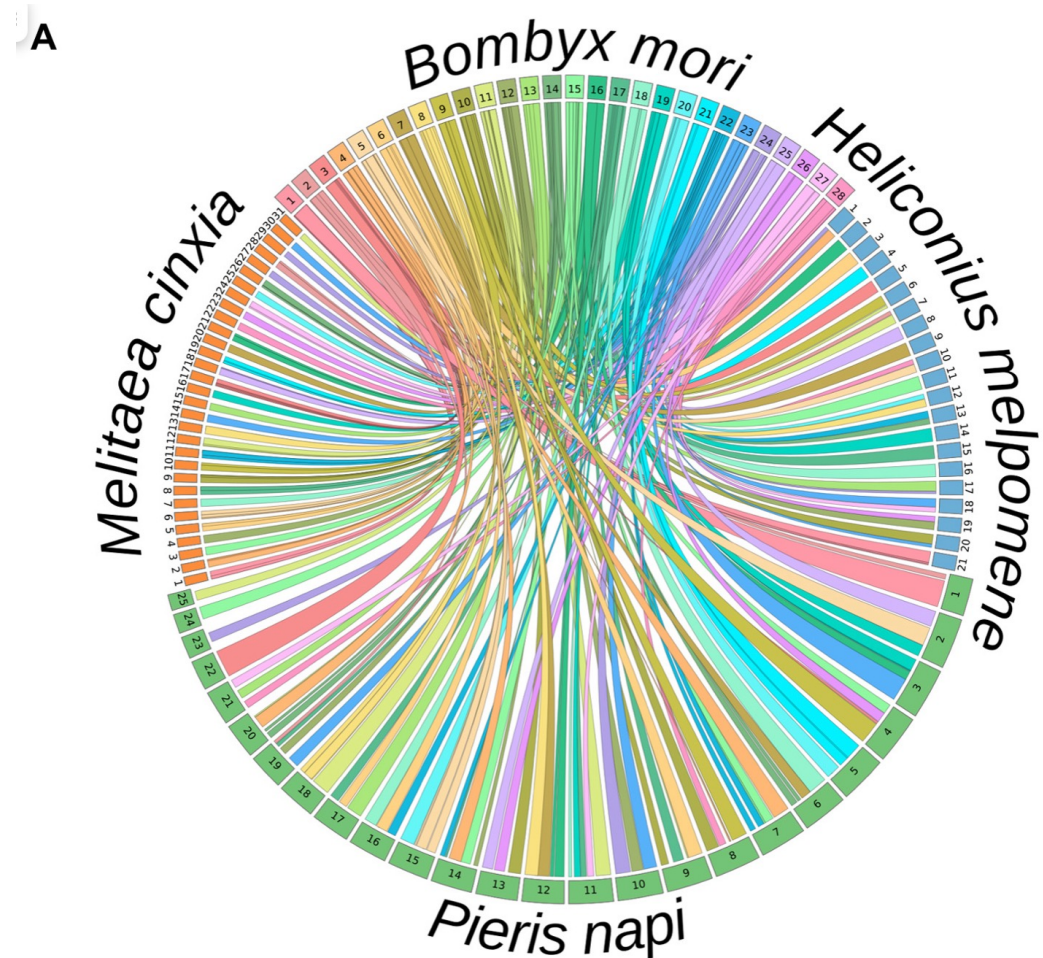
Heterozygosity – proxy for population size?

- Number of segregating sites between the haplotypes
- Runs of homozygosity
 - Long uninterrupted homozygous regions indicate low number of haplotypes segregating in the population
- Caveat: only one individual – is it representative for the population or species?



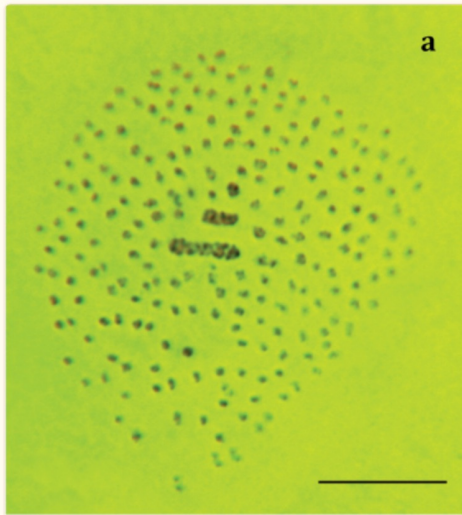
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- Heterozygosity
- **Genome structure - karyotype**



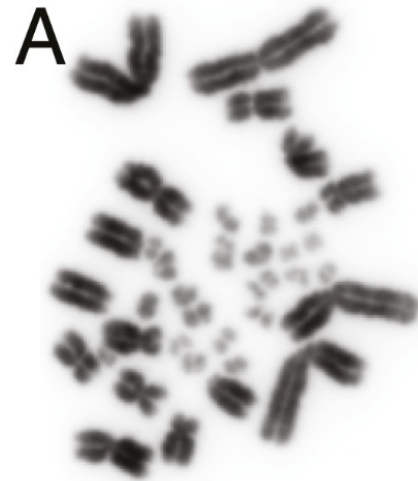
Genome structure - chromosomes

Number



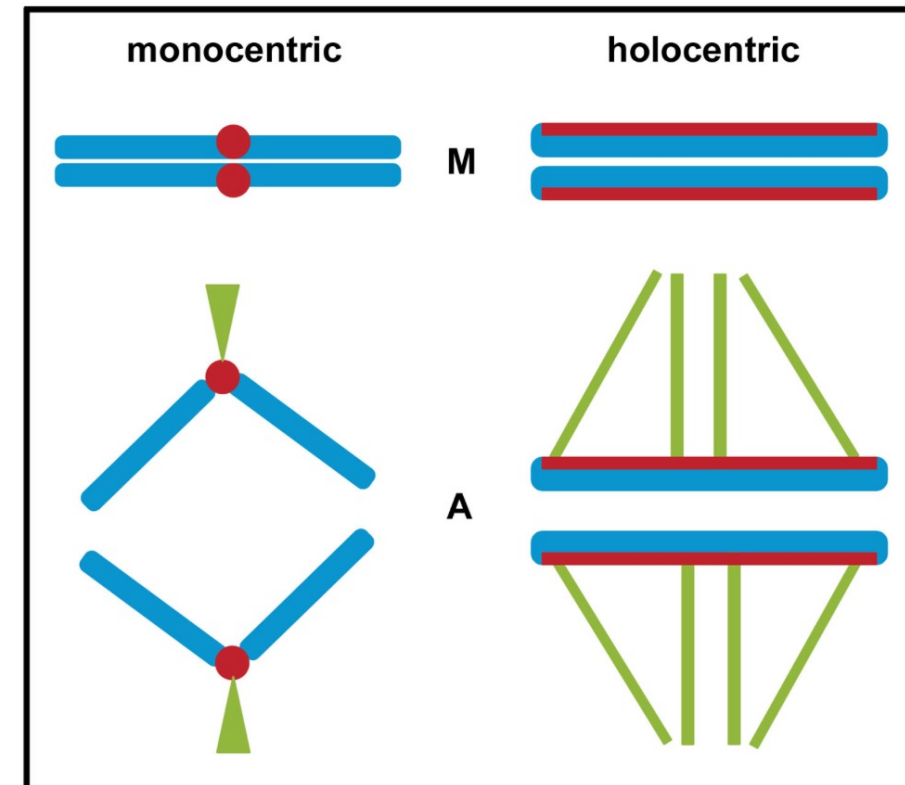
The atlas blue, n=226
Lukhtanov, 2015

Size



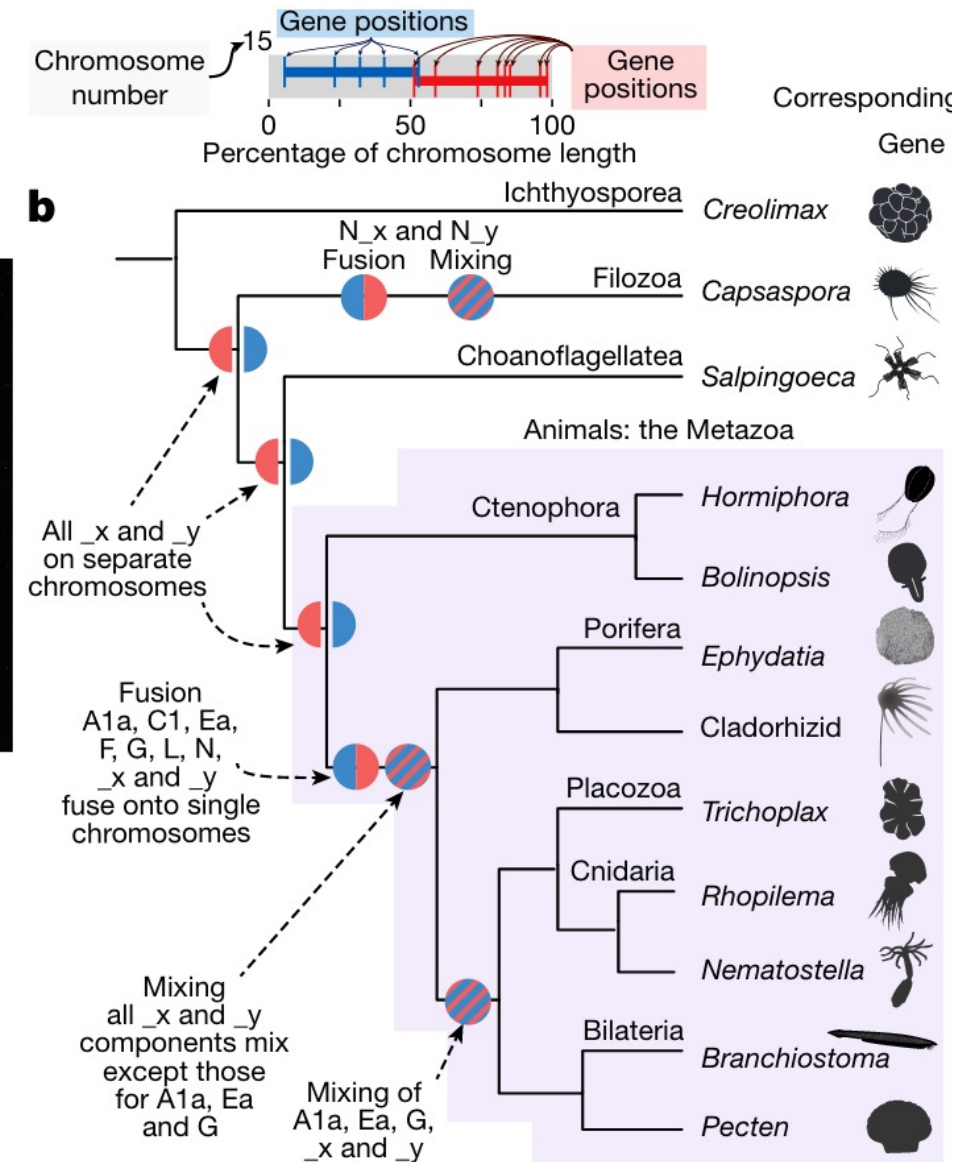
The bearded dragon,
Waters et al 2021

Function and organisation

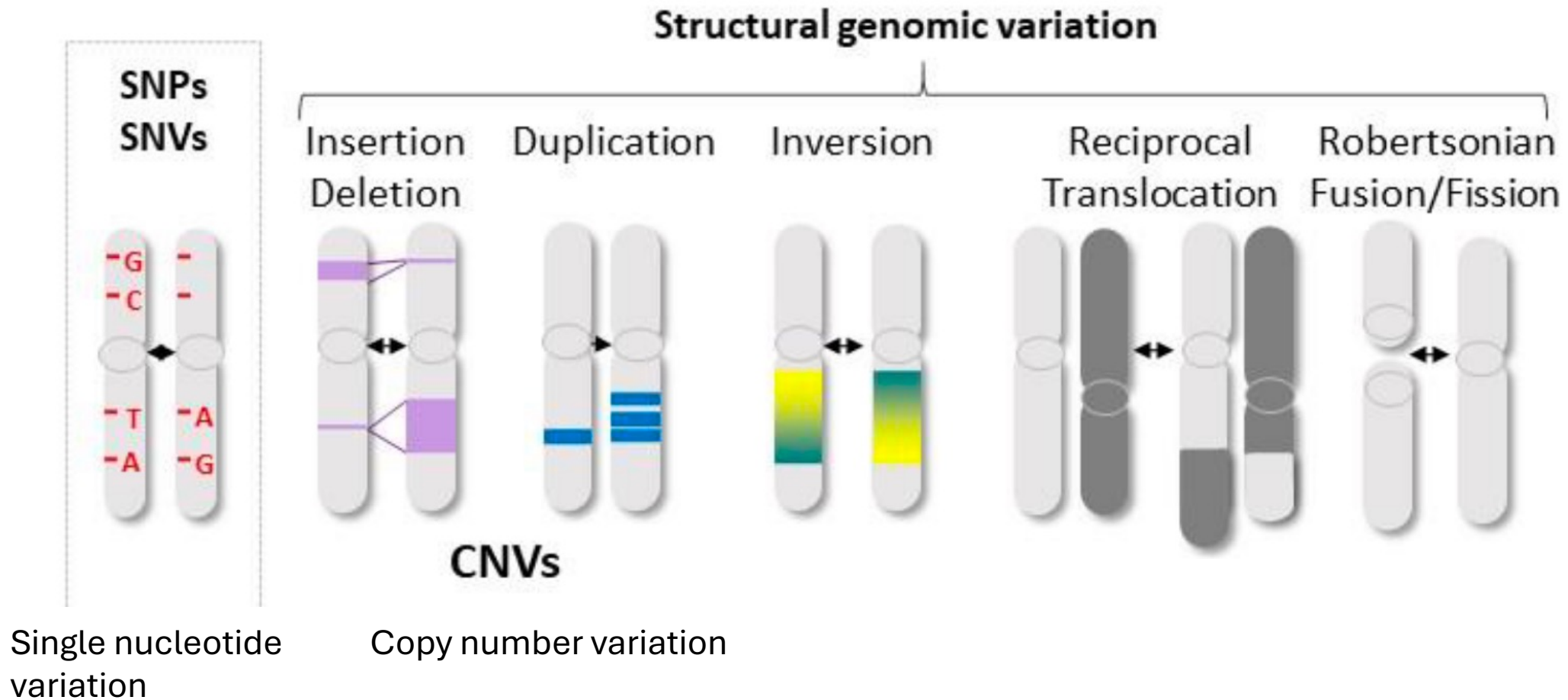


Ancestral reconstruction of karyotype

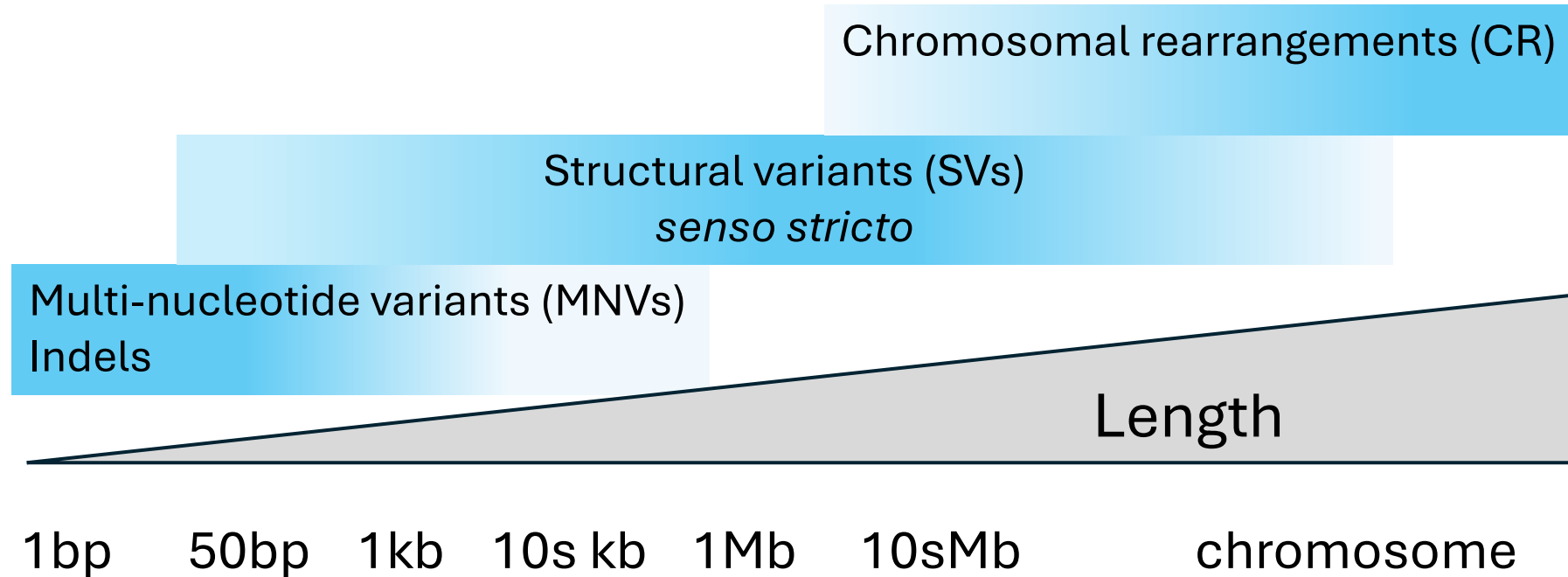
- Infer of rates and patterns of evolutionary change through time
- Resolve phylogenetic relationships
 - Chromosome fusion and subsequent mixing of gene order is an irreversible state
 - Ctenophora likely basal to all other Metazoa



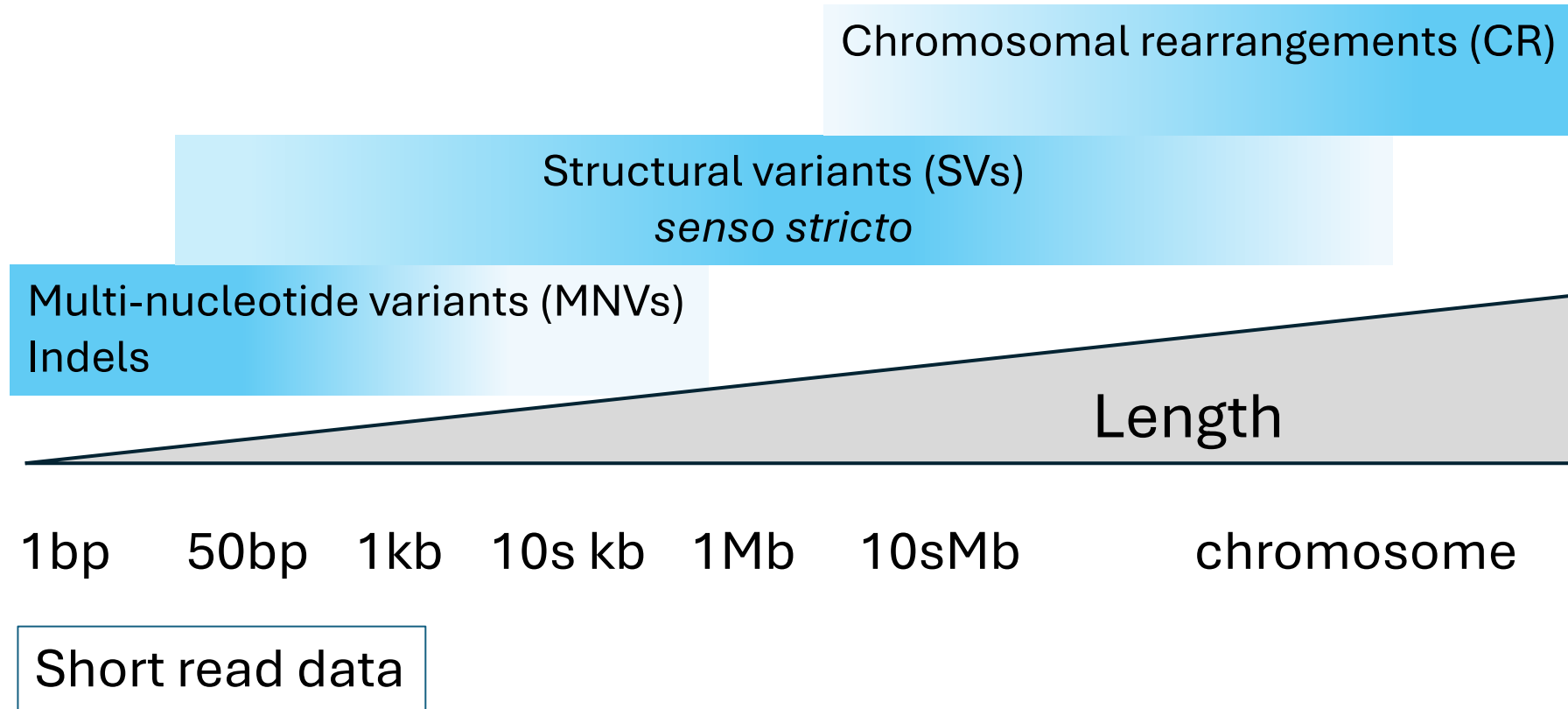
Genome structure – structural variants



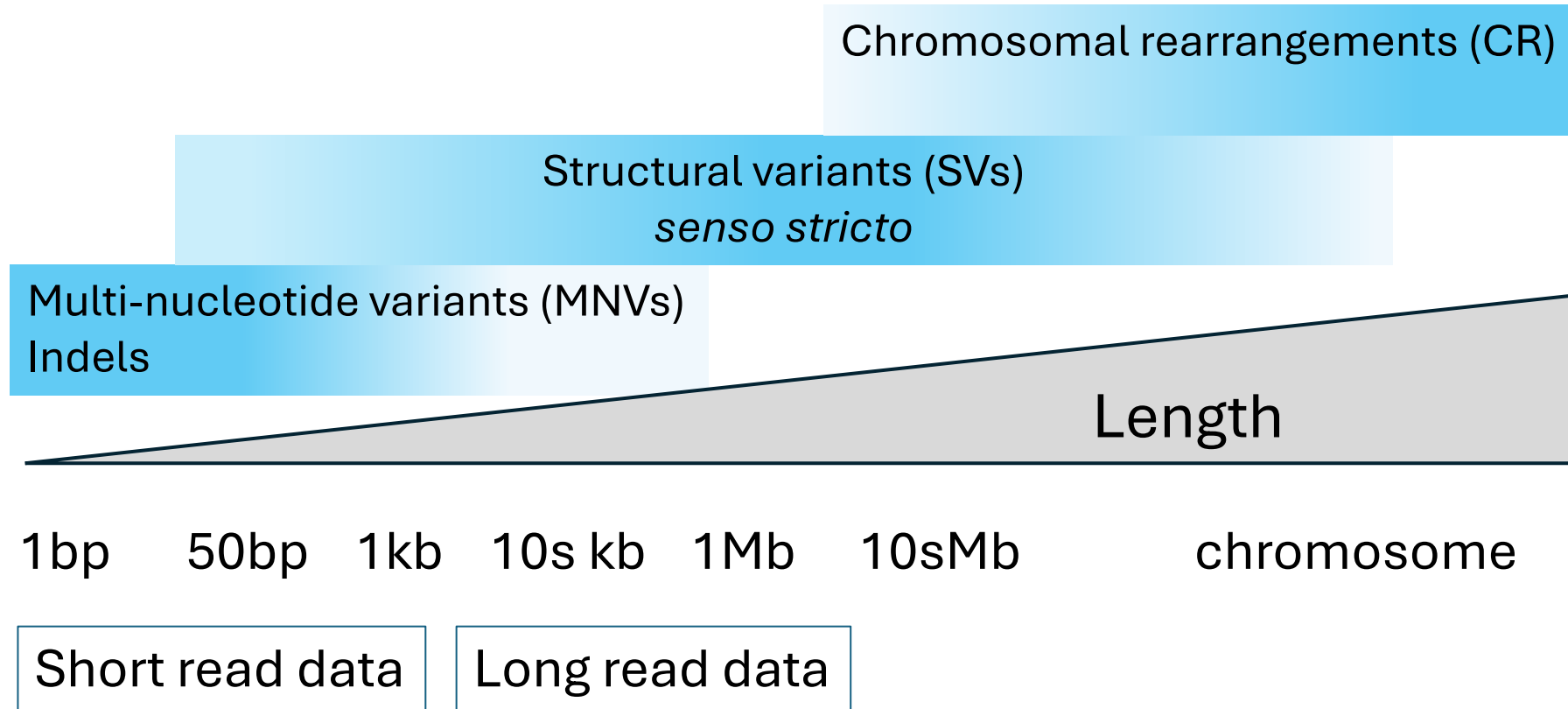
Genome structure – structural variants



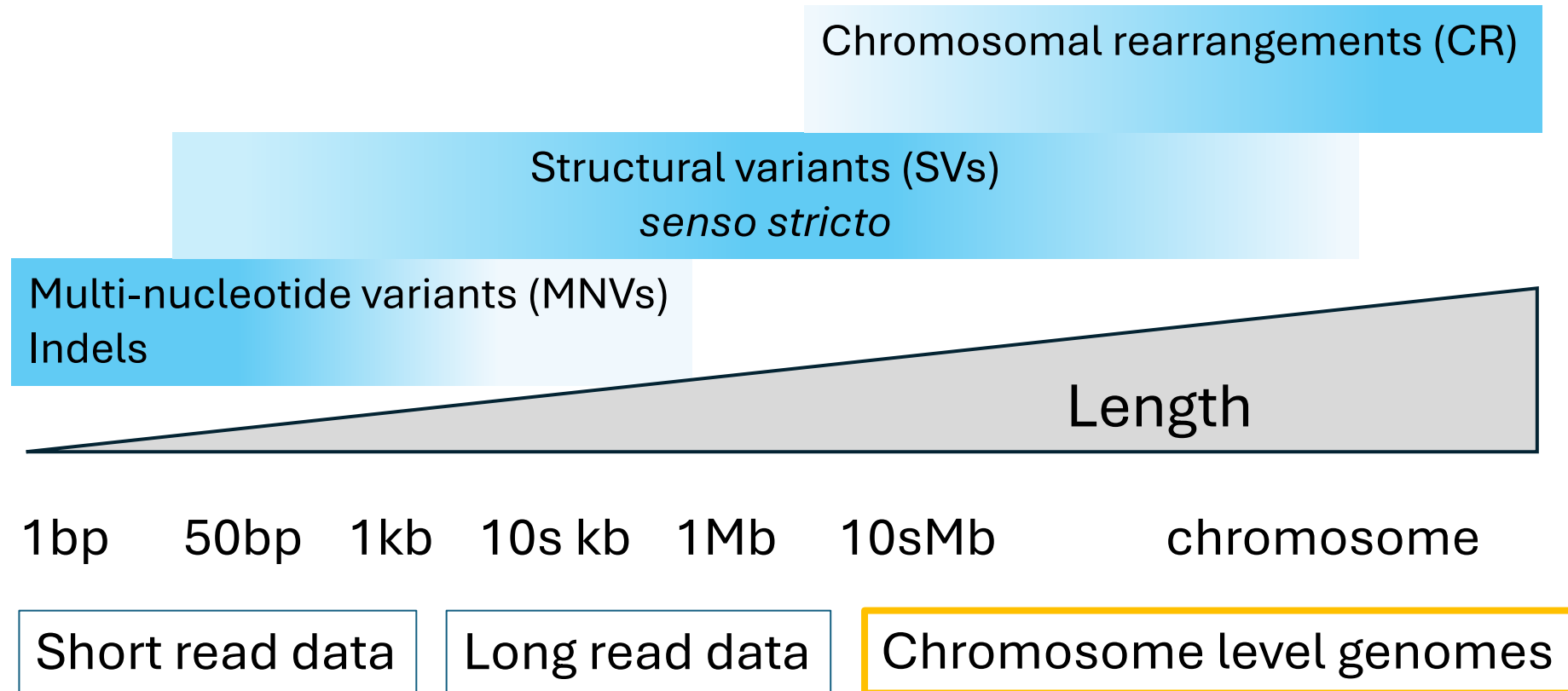
Genome structure – structural variants



Genome structure – structural variants



Genome structure – structural variants



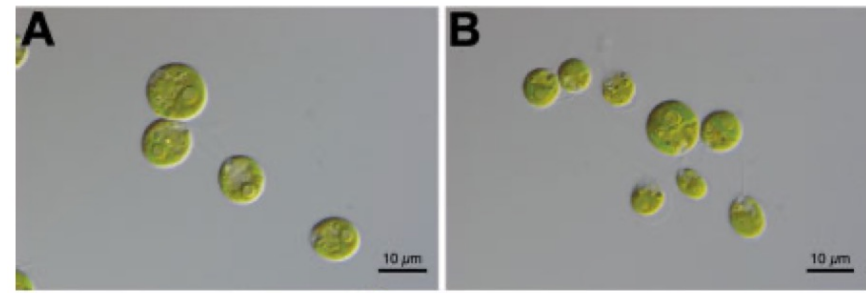
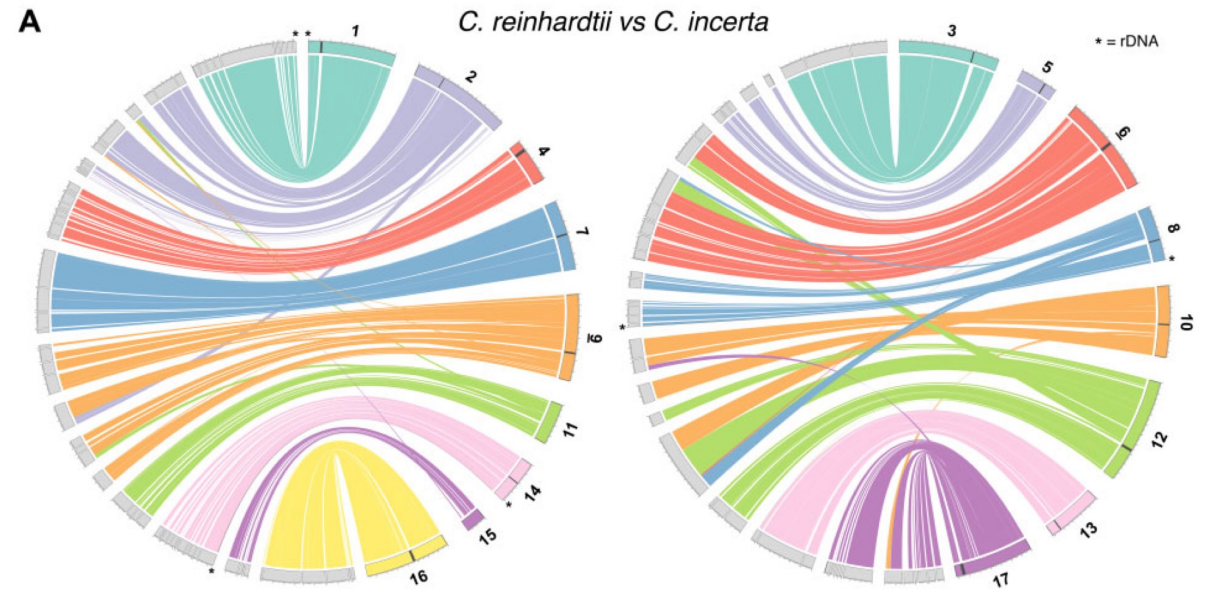
My work

Synteny analysis - exercise

- Whole genome alignment
- Orthologous markers

Whole genome alignment

- Closely related taxa
 - MUMMER
 - Minimap2
 - Cactus
 - LastZ
 - D-Genies

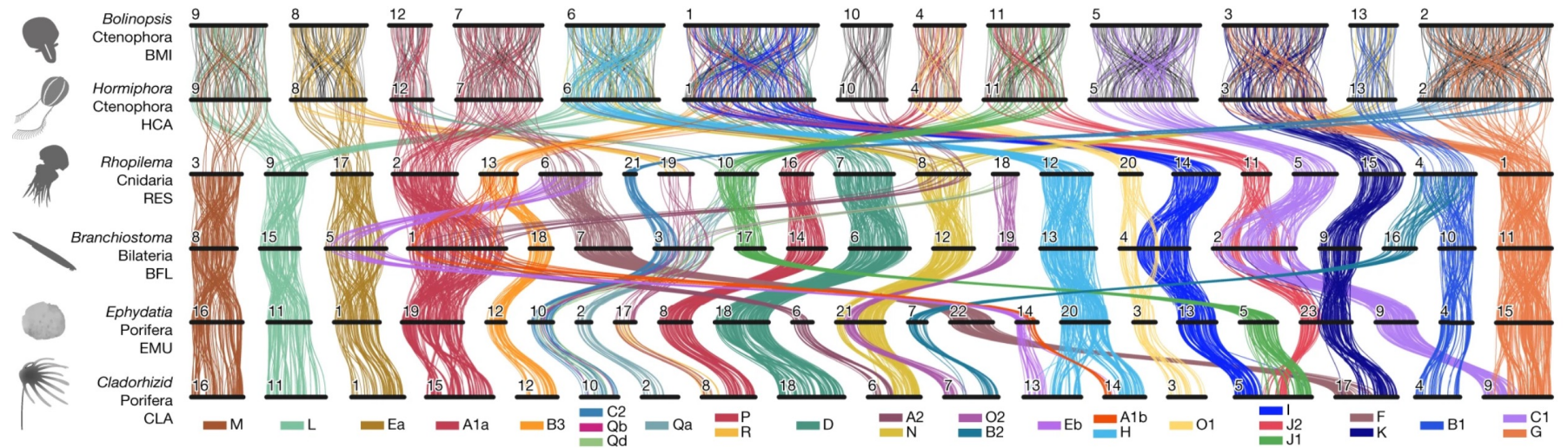


Green algae *Chlamydomonas*

Craig et al 2012

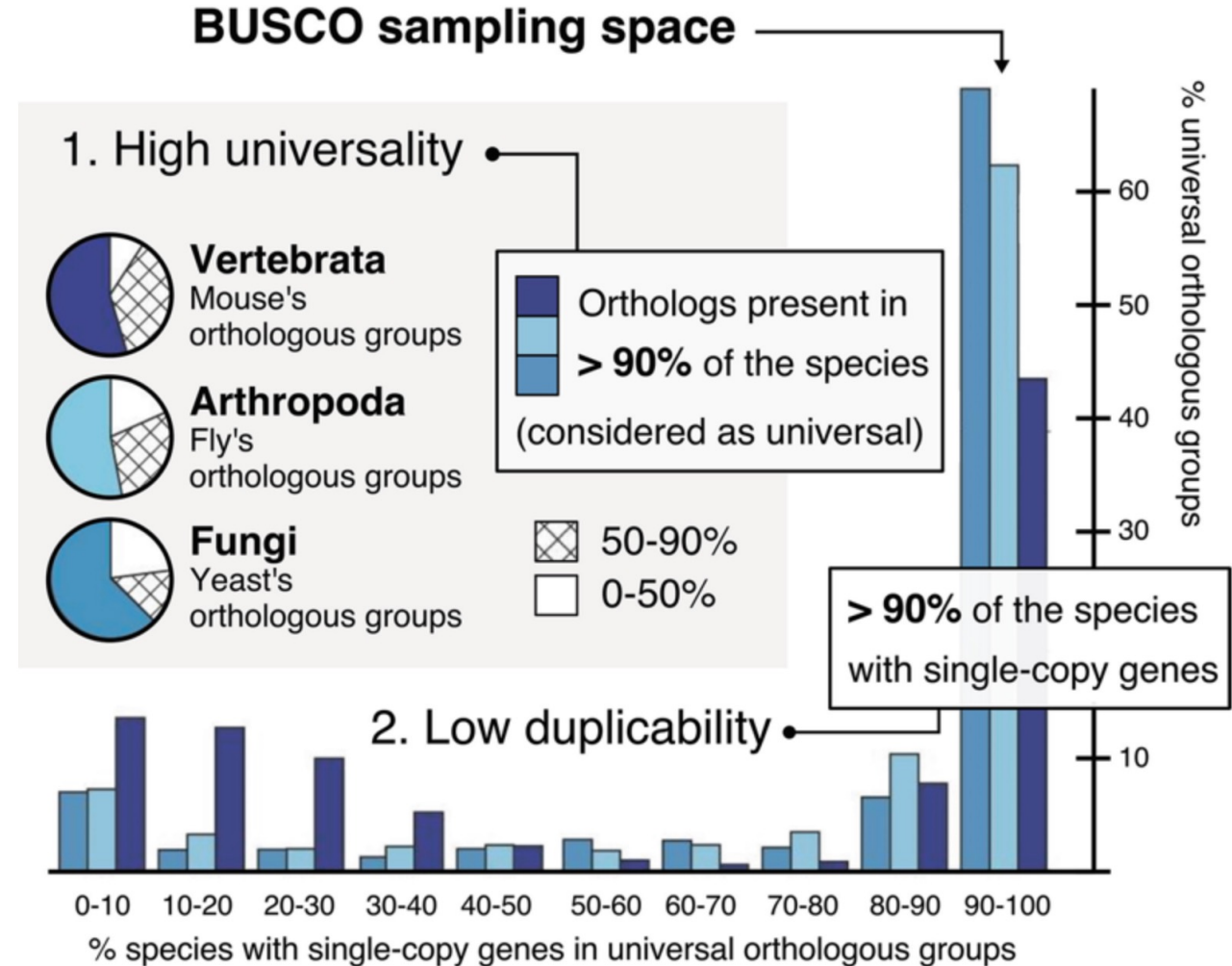
Orthologous markers

- Use conserved sequences as anchor markers
- Also for very divergent taxa

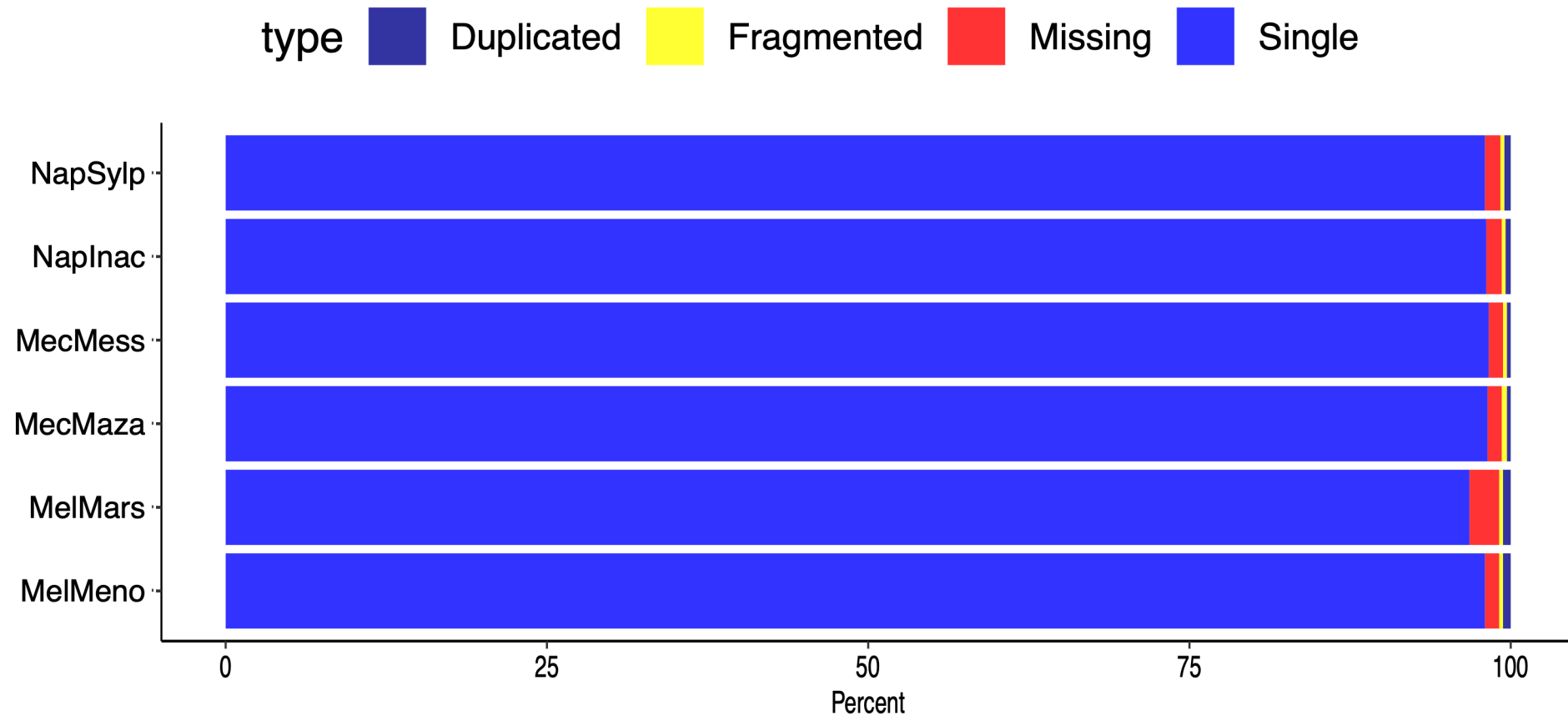


Orthologous markers

- BUSCO (Benchmarking Universal Single Copy Orthologues)
 - BLAST <https://blast.ncbi.nlm.nih.gov/Blast.cgi>
 - Finds regions of similarity between biological sequences. The program compares nucleotide or protein sequences to sequence databases and calculates the statistical significance.

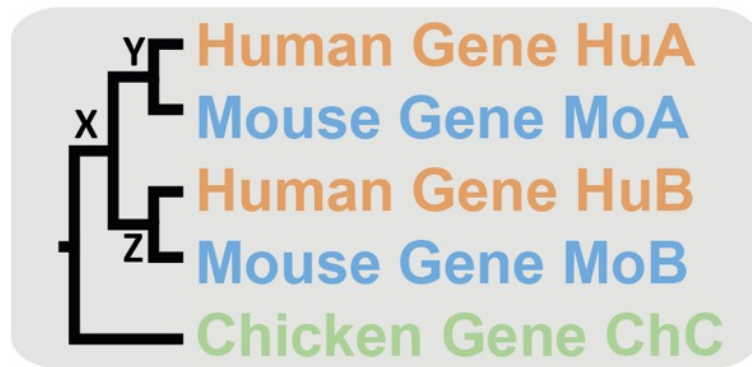


BUSCO completeness



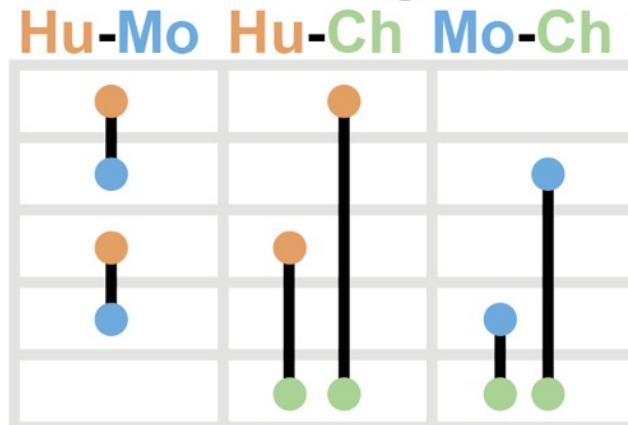
OrthoFinder

A. Orthogroup



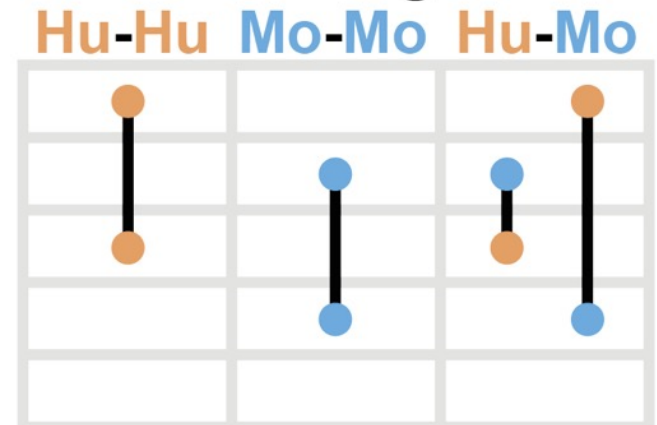
Group of genes descended from single gene in LCA of group of species

B. Orthologs



Pairs of genes descended from single gene in LCA of pair of species

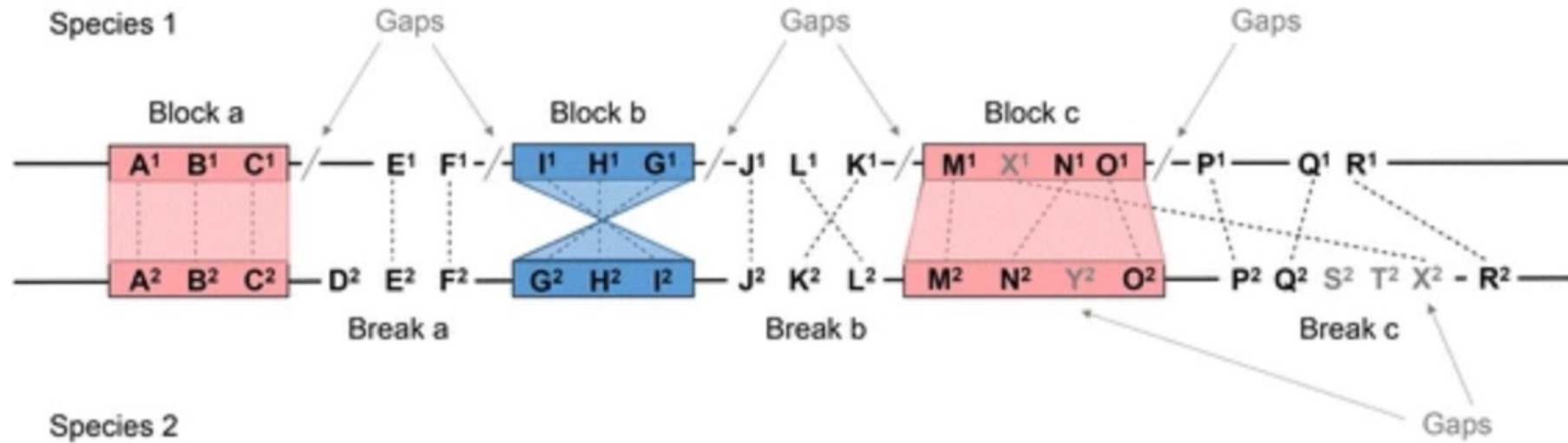
C. Paralogs



Pairs of genes descended from gene duplication event

Visualisation

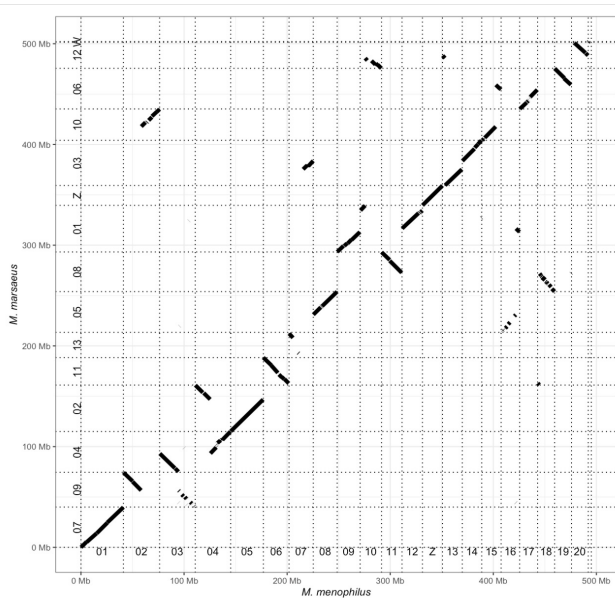
- Synteny



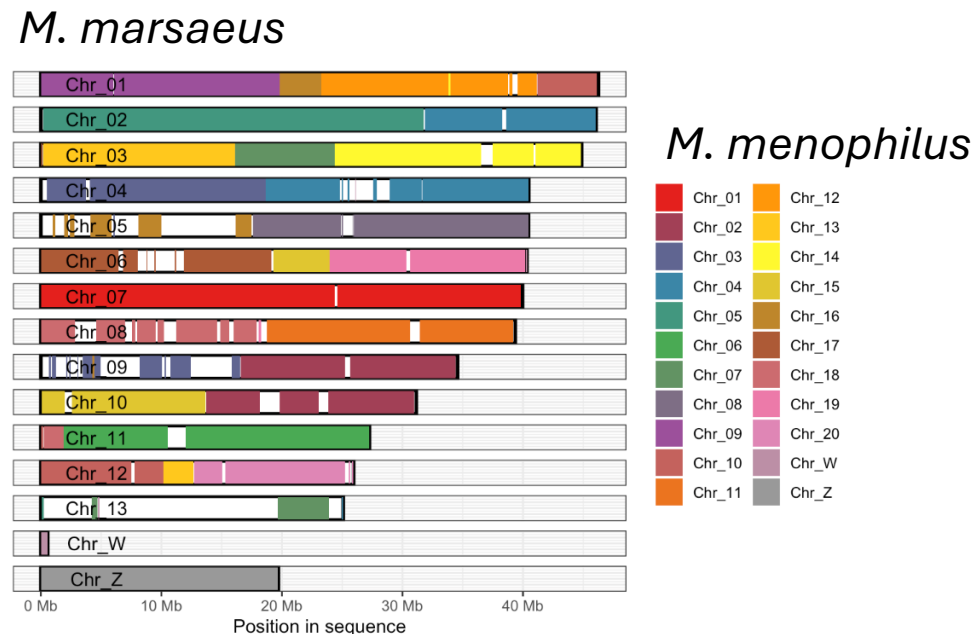
Visualisation

Melinaea marsaeus (n=14) vs *M. menophilus* (n=21)

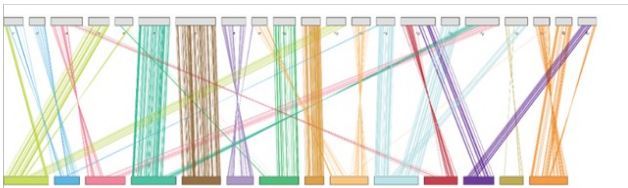
Dotplot



Chromosome painting



Ribbon plot



Hands-on synteny analysis

- Two species of Ithomiini
 - *Mechanitis marsaeus*
 - *Mechanitis messenoides*
- Data type available
 - Reference genomes at NCBI



Hands-on synteny analysis

- Question:
 - Are there any chromosomal rearrangements between the taxa?
- Methods
 - Whole genome alignment
 - MiniMap2
 - Ribbon plot
 - Marker orthology
 - BUSCO
 - Orthofinder
 - Circos
 - Extra: dotplot
<https://dgenies.toulouse.inra.fr>

